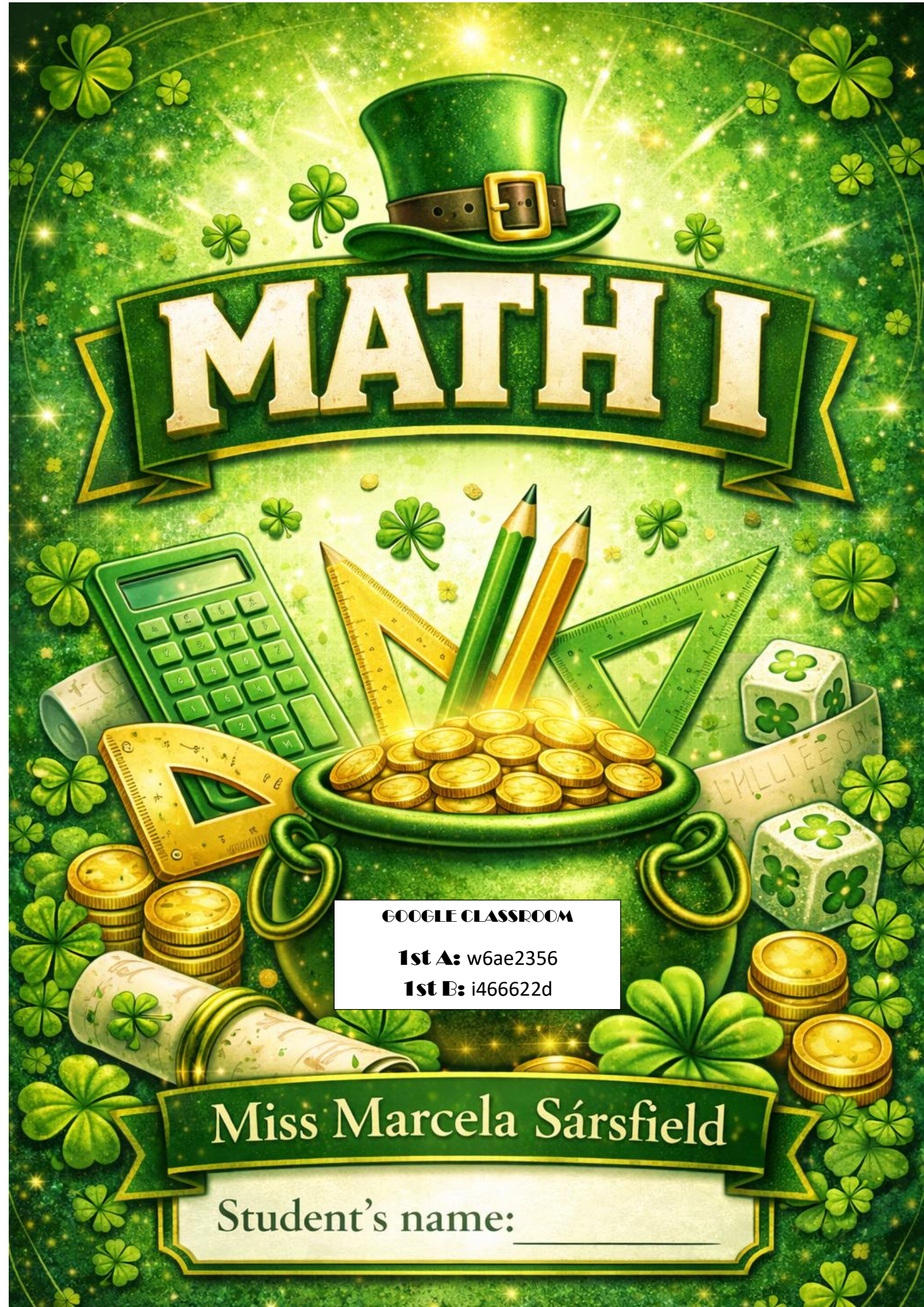

MATH I
GOOGLE CLASSROOM
1st A: w6ae2356
1st B: i466622d
Miss Marcela Sársfield
Student's name:



Subject materials:

- ⊕ Folder with **squared sheets**.
- ⊕ Pencil, color pencils, eraser, protractor and ruler.
- ⊕ Printed booklet.

EVERYONE SHOULD HAVE THE MATERIALS FOR ALL CLASSES.

TESTS

Date	Topics

MATH 1 - 2026

Course: 1st year

Teachers: Marcela Sársfield Arias

Course Objective

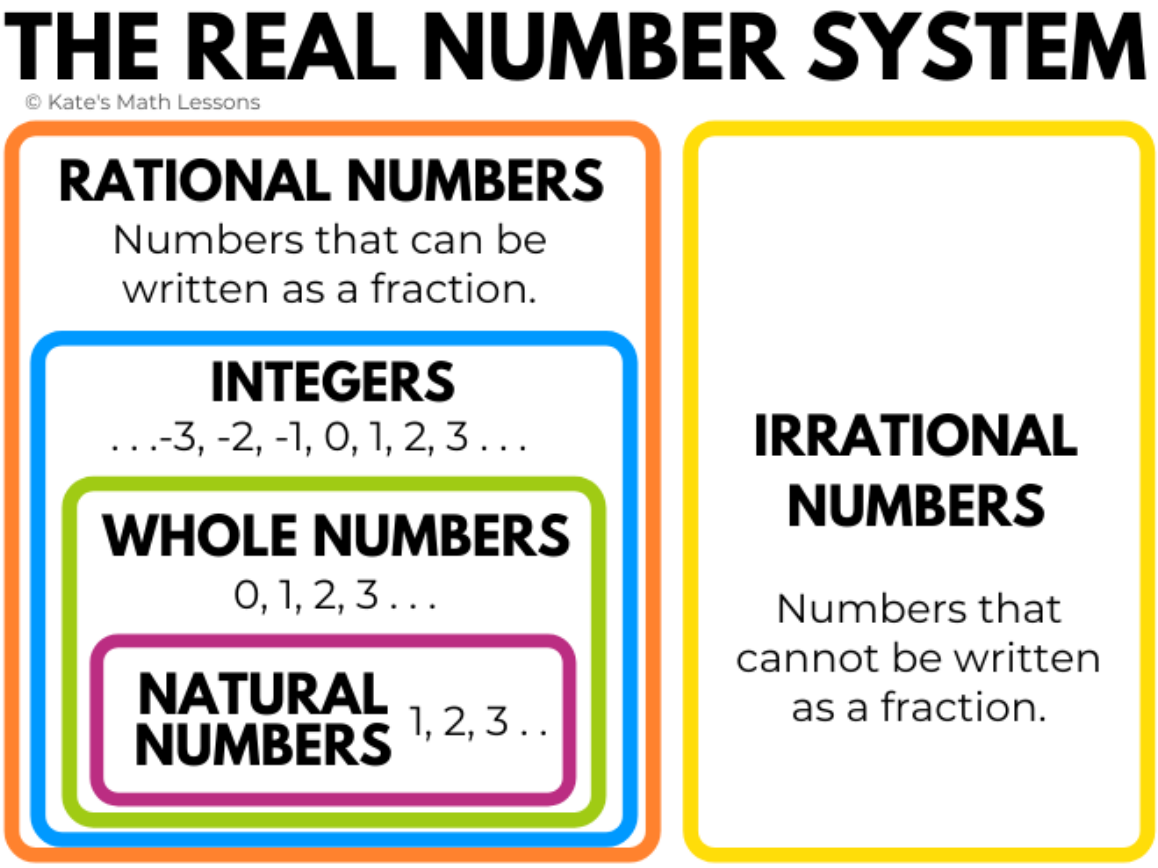
- Get the students prepared to take the S.A.T.
- Understand the significance and meaningful concepts, as well as connections among numeric fields or geometric concepts.
- Be able to transfer to daily life the contents given (including topics, problems and issues related to cross-curricular projects: Sustainable Development Goals and E.S.I.)
- Integrate the knowledge acquired in previous years with the new information
- Promote respect to diversity and equality (E.S.I)

Unit Titles/Topics/Brief Description

Unit 0	**DIAGNOSTIC**
• Natural numbers • Operations with natural numbers. Division. • Fractions. Graph representation. Proper, improper and mixed numbers.	
Unit 1	**NUMBERS AND OPERATIONS**
• Whole numbers. • Operations and Properties of Whole Numbers. • Multiples, factors, and divisors. • Prime and composite numbers. • Divisibility rules. • Prime factorization. GFC. LCM • Exponents. Laws of exponents. • Order of operations (PEMDAS) • Algebraic expressions	
Unit 2	**GEOMETRY AND MEASUREMENTS**
• Points, lines. Types of lines. Number scales. • Two-dimensional figures analysis (Triangles) Characteristics and classification. • Introduction to polygons. • Property of the sum of interior angles of triangles and quadrilaterals.	
Unit 3	**ALGEBRA AND FUNCTIONS**
• Two-dimensional coordinate system. • Locations	
Unit 4	**PROBABILITY AND STATISTICS**
• Data and graphs. Types of data. • Interpretation of tables and graphs.	

DIAGNOSTIC UNIT

Symbol of the Set of Natural Numbers
$\mathbb{N}$
Elements of the Natural Numbers
{ 1, 2, 3, 4, 5, 6, ... }



EVEN/ODD NUMBERS

- **Even Numbers: Any natural number that can be divided exactly by 2.**

The last digit is 0, 2, 4, 6 or 8

Example: 24, 0, 6 and 38 are all even numbers

- **Odd Numbers: Any natural number that cannot be divided exactly by 2.**

The last digit is 1, 3, 5, 7 or 9

Example: −3, 1, 7 and 35 are all odd numbers

NATURAL NUMBERS



Natural Numbers

- It refers to a set of whole numbers excluding zero (0).
- It is used for counting and ordering numbers.
- Natural numbers are infinite (endless).
- The smallest natural number is 1.
- It is written as N= {1, 2, 3, 4, 5, …}

NOT A NATURAL NUMBER

Negative Numbers are numbers smaller than zero
-10, -5, -12

A decimal number is a number that contains a decimal point in which shows the smaller parts of the whole number.
1.34, 8.9, 0.1

Fractions is a representation used to show the smaller parts of a whole
½, ¾, ⅓

Therefore, these numbers are not natural numbers because they are not used in basic counting.

NATURAL NUMBERS CAN BE…

Cardinal Numbers

show the quantity of objects or how many the objects are

Example:
1, 2, 3, 4, 5, …

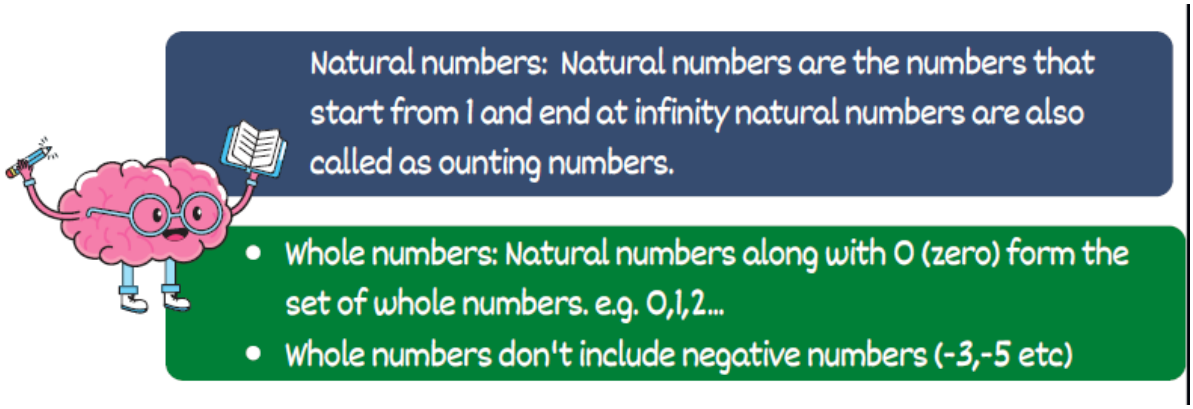
Ordinal Numbers

show the order of how things are set. It can be the ranking of things

Example:
1st, 2nd, 3rd, 4th, 5th, …



WHOLE NUMBERS = NATURAL NUMBERS + ZERO



1. BASIC OPERATIONS:

- **Addition**
- **Subtraction**
- **Multiplication**
  - Signs for Multiplication: "x", "." or just **nothing** between the factors.

$a \times b$	$a \cdot b$	ab

- **Division**

Signs for Division:

$a \div b$	$a : b$	a / b	$\frac{a}{b}$

OPPOSITE OPERATIONS: we use them to "undo".

Addition and subtraction	Multiplication and division
$5 + 2 = 7$ $5 + 2 - \mathbf{2} = 5$	$6 \times 2 = 12$ $6 \times 2 \div \mathbf{2} = 6$

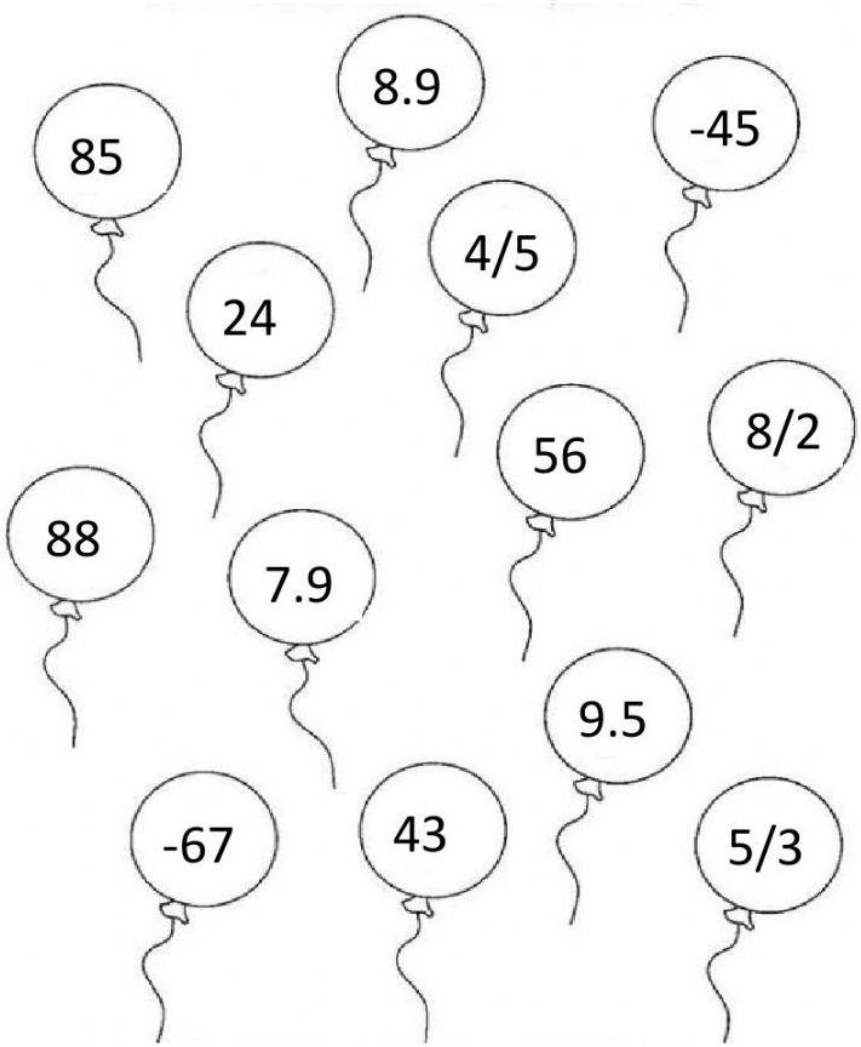
ADDEND + ADDEND = SUM

MINUEND - SUBTRAHEND = DIFFERENCE

FACTOR X FACTOR = PRODUCT

DIVIDEND ÷ DIVISOR = QUOTIENT + REMAINDER

- Color the ballons with natural numbers



1. Write down the **first ten** even numbers.

   Answer =

2. Write down the **even numbers** from 998 to 1 012.

   Answer =

3. Write down the **first twelve** odd numbers.

   Answer =

4. Write down the **odd numbers** from 1 003 to 1 013.

   Answer =

Look carefully at the numbers below.

Put a **circle** around all the **even** numbers.

Put a **rectangle** around all the **odd** numbers.

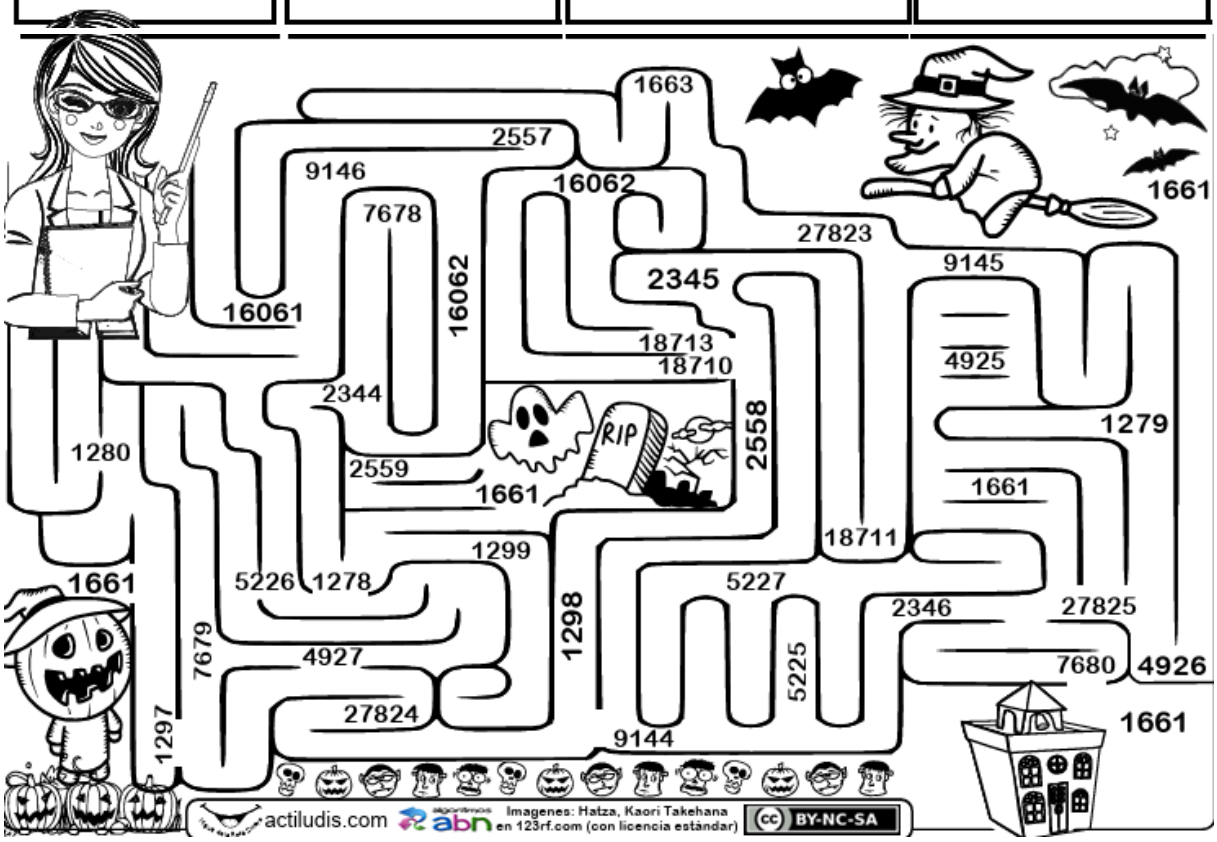
Two examples have been done for you.



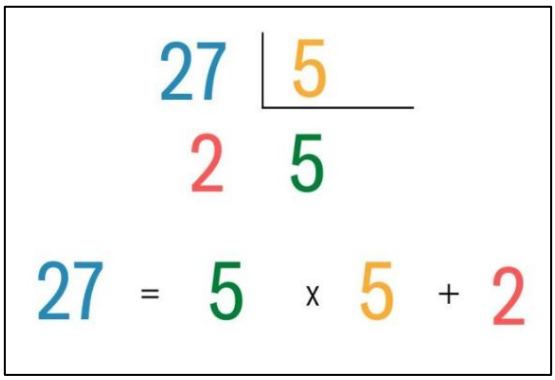
34 089	6 998	54	156 110
1 053	2 022	46 095	779
7 991	217	809 116	1 678 322

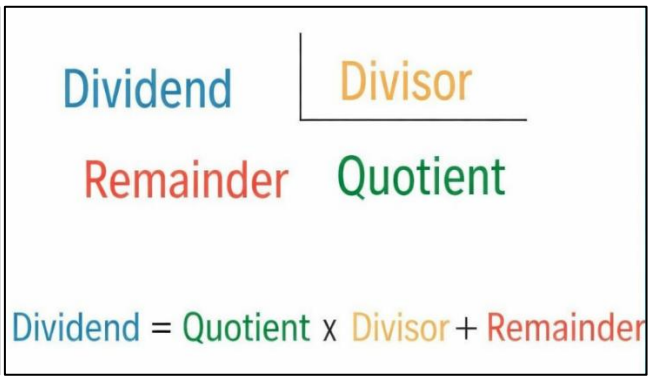
- **Perform the operations and look them up in order to find the path.**

1 6874+ 9187	2 7605 - 2379	3 4786+ 436+ 2457	4 3478 X 8
5 9087 : 7	6 9827 - 7269	7 2568+ 652 - 875	8 2673 X 7
9 5678+ 3467	10 3457 - 2178	11 6778 - 895 - 957	12 9970 : 6



DIVISION





- **The remainder is always less than the divisor.**
- **If the remainder is 0, the division is exact**

* Complete the following charts

DIVIDEND	DIVISOR	QUOTIENT	REMAINDER
37	4	9	1
58	6		
253	12		
293	9		
3.184	15		
70	10		

Now solve the others:

DIVIDEND	DIVISOR	QUOTIENT	REMAINDER
40		3	
297		21	3
500		50	0
608	14		6
260		20	

* Complete:

▢ | 5 4 3	▢ | 9 2 2	▢ | 2 1 6
▢ | 9 6 7	▢ | 4 3 5	▢ | 8 7 2

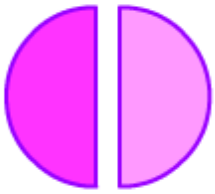
* If the divisor is 7, what are all the possible remainders? Why?

FRACTIONS

A fraction is any part of a group, number or whole.



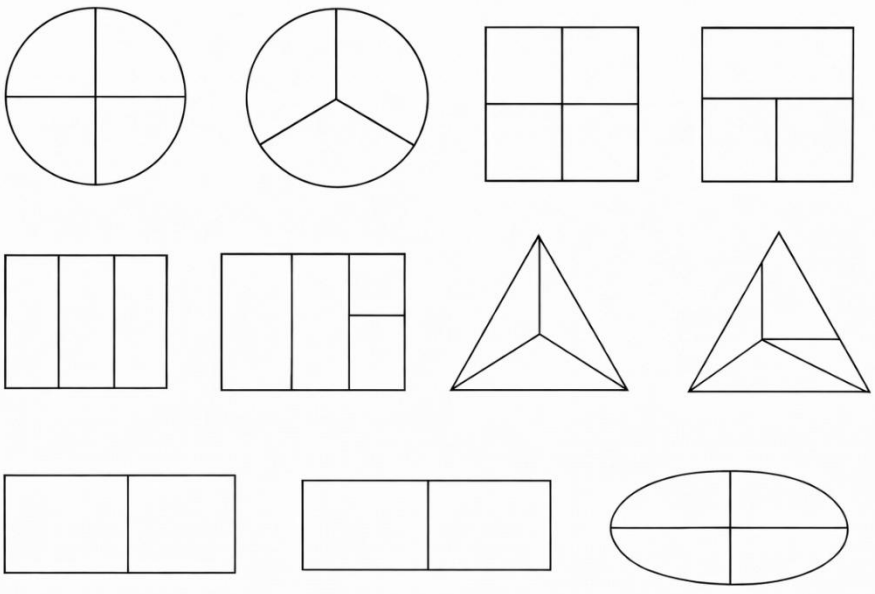
One circle has been cut in half.
A half is a fraction.
We write one half as



The top number is called the numerator. **1** **It is the number of parts we have.**

The bottom number is called the denominator. **2** **It is the total number of parts the whole is divided into.**

Which diagrams represent fractions?



There are three main types of fractions.

proper fraction

$\frac{1}{2}$ 1 numerator, 2 denominator

The numerator is less than the denominator.



$\frac{1}{4}$



$\frac{2}{3}$



$\frac{7}{10}$

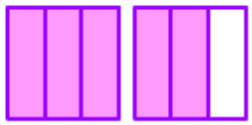
improper fraction

$\frac{5}{2}$ 5 numerator, 2 denominator

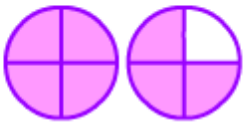
The numerator is larger than or equal to the denominator.



$\frac{4}{4}$



$\frac{5}{3}$



$\frac{7}{4}$

mixed number

$2\frac{1}{2}$

A number written as a whole number with a proper fraction.



$2\frac{2}{3}$



$2\frac{3}{4}$

© Jenny Eather 2014

A) Identify whether the following fractions are proper or improper.

1) $\frac{2}{3}$
   a) proper
   b) improper

2) $\frac{3}{11}$
   a) improper
   b) proper

3) $\frac{5}{4}$
   a) improper
   b) proper

B) Match the following.

$\frac{3}{9}$	• proper •	$\frac{7}{6}$
$\frac{8}{5}$		$\frac{4}{7}$
$\frac{4}{10}$	• improper •	$\frac{9}{4}$

C) 1) Which of the following is a proper fraction?

a) $\frac{6}{7}$ b) $\frac{3}{2}$ c) $\frac{9}{5}$ d) $\frac{7}{3}$

2) Which of the following is not a proper fraction?

a) $\frac{2}{5}$ b) $\frac{8}{3}$ c) $\frac{3}{8}$ d) $\frac{1}{9}$

3) Which of the following is an improper fraction?

a) $\frac{1}{5}$ b) $\frac{1}{7}$ c) $\frac{6}{5}$ d) $\frac{6}{11}$

* Represent these fractions:

1. 3/4
2. 5/8

3. 7/3
4. 9/5
5. 2/6
6. 11/4

* Which fractions are **proper fractions**?
* Which fractions are **improper fractions**?

UNIT 1: NUMBERS AND OPERATIONS

What are the Properties of Whole Numbers?

Whole numbers are countable numbers that include zero and are generally used for calculation purposes. The set of whole numbers in Mathematics starts from 0, that is {0, 1, 2, 3, ...}. The set of whole numbers is denoted by the symbol **W**.
The properties of whole numbers are as follows:

- Closure Property
- Associative Property
- Commutative Property
- Distributive Property
- Identity Property

PROPERTIES OF OPERATIONS WITH WHOLE NUMBERS

- **Addition**

Closure	Associative	Commutative	Identity
The sum of **two natural numbers** is also a **natural number**. **a + b**	The way in which the addends are grouped does not change the result. **(a + b) + c = a + (b + c)** (2+3) + 5 = 2 + (3+5) 5 + 5 = 2 + 8 10 = 10	The order of the addends does not change the addition. **a + b = b + a** 2 + 5 = 5 + 2 7 = 7	The sum of any natural number and zero is the natural number itself. **a + 0 = a** 6+0=0+6=6 Zero (0) is the additive identity for natural numbers.

- **Subtraction**

No closure	No commutative	Identity
The result of **subtracting two natural numbers** is not always another **natural number**. 2 − 5 ∉N	5 − 2 ≠ 2 − 5	The subtraction of any natural number and zero is the natural number itself. **a + 0 = a** 6+0=0+6=6 Zero (0) is the additive identity for natural numbers.

- **Multiplication**

Closure	The result of multiplying **two natural numbers** is another **natural number**. **a · b** ∈ N
Associative	The way in which the factors are grouped does not change the result. **(a · b) · c = a · (b · c)** (2 · 3) · 5 = 2 · (3 · 5) 6 · 5 = 2 · 15 30 = 30
Commutative	The order of the factors does not change the product. **a · b = b · a** 2 · 5 = 5 · 2 10 = 10
Multiplicative Identity	The **1** is the neutral element of the multiplication because any number multiplied by it gives the same number. **a · 1 = a** 3 · 1 = 3
Distributive	The multiplication of a natural number and a sum is equal to the sum of the **multiplication** of the **natural number** for each of the **addends**. **a · (b + c) = a · b + a · c** 2 · (3 + 5) = 2 · 3 + 2 · 5 2 · 8 = 6 + 10 16 = 16 **Removing a common factor:** It is the reverse of the distributive property. **a · b + a · c = a · (b + c)** 2 · 3 + 2 · 5 = 2 · (3 + 5) 6 + 10 = 2 · 8 16 = 16

- **Division**

No closure	**No commutative**	**Identity**
The result of dividing **two natural numbers** is not always another **natural number**. 2: 6 ∉N	**a: b ≠ b: a** 6:2 ≠ 2:6	If a number is divided by 1, the result is the same number. **a : 1 = a** **8 : 1 = 8**

Properties of operation on whole numbers

	Addition	Subtraction	Multiplication	Division
Closure Property	✓	✓	✓	✗
Commutative Property	✓	✗	✓	✗
Associative Property	✓	✗	✓	✗

1. Choose the correct option.

A. 7 + 4 = 4 + 7, Identify the property

a. commutative b. closure c. property of zero d. Associative

B. (a + b) + c = a + (b + c), identify the property.

a. commutative b. closure c. property of zero d. Associative

C. The smallest natural number is:

a. 0 b. 1 c. 2 d. 10

D. Which of the following natural number does not have a predecessor?

a. 0 b. 1 c. 2 d. 999999999

E. In which of the following operation, all three properties i.e closure, commutative and associative will not work?

a. Addition b. Multiplication and division

b. Division only d. Division and subtraction

F. Identify the property: $a \times 0 = 0 \times a = 0$

a. closure b. property of zero

c. multiplicative identity d. Associative

G. Given two whole numbers a and b, which of the following may not always be whole numbers?

a. $a + b$ b. $a - b$ c. $a \times b$ d. $2a + b$

H. Which whole number x does not satisfy the relation $x \div x = 1$

a. 0 b. 2 c. 25 d. 50



- Zero is the smallest whole number
- 1 is the smallest natural number
- All the natural numbers are whole numbers, but all whole numbers are not natural numbers

2. Identify the property used.

a. $56 + 12 = 12 + 56$ _____

b. $19 + 0 = 19$ _____

c. $(45 + 12) + 8 = 45 + (12 + 8)$ _____

d. $10 \times 1 = 1 \times 10 = 10$ _____

3. Find the successor and predecessor of:

a. 5698 b. 101010 c. 99999

4. Fill in the blanks.

a. 504 + ___ = 112 + 504
b. 101 + 0 = ___ + 101
c. 90 + 43 + 10 + 7 = 100 + ____
d. 112 + (56 + 12) = (___ + 56) + 12

5. Which of the following statement is false? cross it and justify your answer.

a. 16 - 2 = 2 - 16 _____
b. 88 + 4 = 4 + 88 _____
c. 100 - 0 = 0 - 100 _____
d. (19 - 4) - 1 = 19 - (4 - 1) _____
e. 99 x 100 = 100 x 99 _____
f. 20 ÷ 5 = 5 ÷ 20 _____

6. Solve the following using the properties only.

a. 465 + 12 + 45 b. 796 + 245 + 204
c. 25 x 896 x 4 d. 4 x 16 x 25

7. Show that:

a. (865 + 452) + 60 = 865 + (452 + 60)
b. (15 x 20) x 2 = 15 x (20 x 2)
c. 99 + (101 + 102) = (99 + 101) + 102



- Distributive property of addition over multiplication
- a x (b + c) = a x b + a x c

e.g. 9 x (6 + 2) = 9 x 6 + 9 x 2
= 54 + 18 = 72

- Distributive property also work for subtraction
- a x (b - c) = a x b - a x c

- solving using distributive property

12 x 99 = 12 (100 -1) { 99 can be written as 100 -1}

12 x 100 - 12 x 1 = 1200 - 12 = 1,188

7 x 102 = 7 x (100 + 2) { 102 can be written as 100 + 2}

8 . Fill in the blanks.

a. 27 x 6 + 27 x 9 = 27 (6 + _ _)
b. 52 x 4 - 52 x 3 = _ _ _ (4 - 3)
c. 22 x 12 + 22 x 14 = 22 x _ _ _ _ _ _
d. 5 x 107 = 5 x(100 + _ _)
e. 12 x 98 = 12 x (_ _ _ - 2)

9. Solve uing distributive property only.

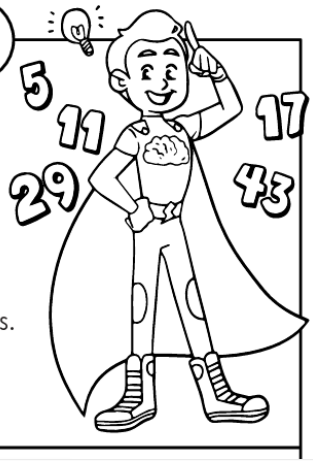
a. 5 x 107 b. 12 x 98 c. 18 x 102 d. 30 x 99
e. 30 x 5 + 40 x 5 f. 15 x 12 - 5 x 12 g. 7 x 6 - 7 x 4

Some tips when operating with natural numbers:

- The product of any number multiplied by 1 is equal to the same number (multiplicand).
- The product of any number multiplied by 0 is equal to 0.
- The quotient of any number divided by 1 is equal to the same number (dividend).
- If we divide 0 by any number, the quotient is 0.
- The quotient of any number divided by 0, the operation is NOT DEFINED.

PRIME AND COMPOSITE NUMBERS

Prime and Composite



Factors are the numbers you multiply to get another number.

Prime numbers are the numbers that have only two factors.
What are the factors of 3? 1 and 3
Because 3 has only two factors, it is a prime number.

Composite numbers are the numbers that have more than two factors.
What are the factors of 6? 1, 2, 3, and 6
Because 6 has more than two factors, it is a composite number.

Complete the table.

Number	List of Factors	Prime or Composite
10		
5		
12		
18		
41		
15		

SIEVE OF ERATOSTHENES TO 100

- Start at number 2. Put a circle round the number.
- Cross out all the multiples of that number.
- Move on to the next number you have not crossed out (3). Put a circle round the number.
- Cross out all the multiples of that number.
- Move on to the next number not crossed out (5). Repeat.



Watch the following video:
https://www.youtube.com/watch?v=V08g_lkKj6Q

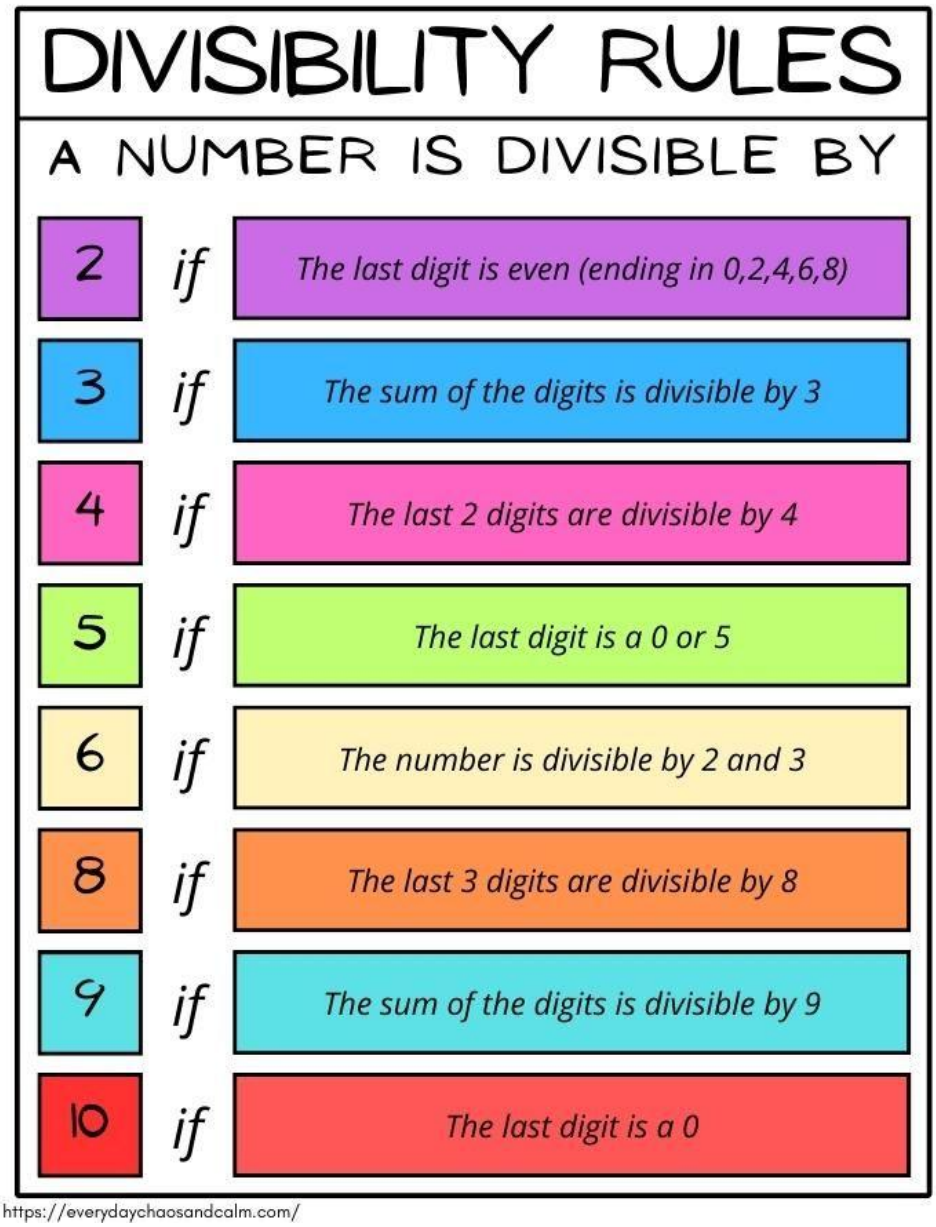
	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

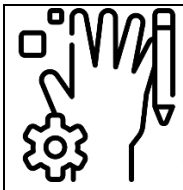
DIVISIBILITY

A number ***a*** is ***divisible*** by another number ***b*** if the division ***a*** ÷ ***b*** is exact (no remainder).

For example, 18 ÷ 3 = 6. So, 18 is divisible by 3. Also, 18 is divisible by 6, because we can write the other division 18 ÷ 6 = 3. So, 18 is divisible by both 6 and 3.

We say 6 and 3 are ***divisors*** or ***factors*** of 18.





Divisibility rule foldable:

https://aharri5on.wordpress.com/2016/02/18/divisibility-rules-foldable/

Factors and multiples are different things. But they both involve multiplication.

Factors

"Factors" are the numbers we can **multiply together** to get another number:

$$2 \times 3 = 6$$

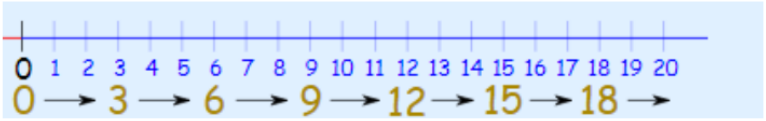
Factor Factor

2 and 3 are factors of 6

Multiples

A multiple is the result of multiplying a number by another number (not a fraction).

Example: Multiples of 3:



Example: 15 **is** a multiple of 3, as 3 × 5 = 15
Example: 16 is **not** a multiple of 3

Factors		Multiples								
1	**1**	2	3	4	5	6	7	8	9	10
1, 2	**2**	4	6	8	10	12	14	16	18	20
1, 3	**3**	6	9	12	15	18	21	24	27	30
1, 2, 4	**4**	8	12	16	20	24	28	32	36	40
1, 5	**5**	10	15	20	25	30	35	40	45	50
1, 2, 3, 6	**6**	12	18	24	30	36	42	48	54	60
1, 7	**7**	14	21	28	35	42	49	56	63	70
1, 2, 4, 8	**8**	16	24	32	40	48	56	64	72	80
1, 3, 9	**9**	18	27	36	45	54	63	72	81	90
1, 2, 5, 10	**10**	20	30	40	50	60	70	80	90	100

EXERCISES

1. Complete with “a multiple”, “a factor” or “divisible”.

   a) 36 is _____ of 4 and 9

   b) 5 is _____ of 20

   c) 40 is _____ by 5

   d) 8 is _____ by 2 and 4

   e) 92 is _____ of 2

   f) 3 is _____ of 45

   g) 240 is _____ of 6 because it is _____ by 2 and 3

2. True or false? Justify your answer.

   a) 35 is a multiple of 5
   b) 8 is a multiple of 56
   c) 20 is divisible by 4
   d) 18 is a divisor of 3
   e) 49 is a multiple of 6
   f) 12 is a factor of 48

23	65	92	10	104	70	500	21	64	798	1223

1 Look at the box of numbers above. Which of these numbers are:

a divisible by 5? **b** divisible by 8? **c** divisible by 3?

2 Say whether the following are true or false.

a 625 is divisible by 5 **b** 88 is divisible by 3
c 640 is divisible by 6 **d** 346 is divisible by 4
e 476 is divisible by 8 **f** 2340 is divisible by 9
g 2890 is divisible by 6 **h** 4562 is divisible by 3
i 40 090 is divisible by 5 **j** 123 456 is divisible by 9

3 Can \$34.07 be divided equally among:

a two people? **b** three people? **c** nine people?

4 A stadium has 202 008 seats. Can these be divided equally into:

a five blocks? **b** six blocks? **c** nine blocks?

5 **a** If a number is divisible by 12, what other numbers must it be divisible by?
b If a number is divisible by 36, what other numbers must it be divisible by?
c How could you test if a number is divisible by 12, 15 or 24?

6 Jacqueline and Sophia stand facing one another. At exactly the same moment both girls start to turn steadily on the spot.
It takes Jacqueline 3 seconds to complete one full turn, whilst Sophia takes 4 seconds to make on full turn.
How many times will Jacqueline have turned when the girls are next facing each other?

E

What is a "Common Factor" ?

Say we have worked out the factors of two numbers:

Example: Factors of 12 and 30

Factors of 12 are **1, 2, 3, 4, 6** and **12**

Factors of 30 are **1, 2, 3, 5, 6, 10, 15** and **30**

Then the **common factors** are those that are found in both lists:

- Notice that **1, 2, 3** and **6** appear in both lists?
- So, the **common factors** of 12 and 30 are: **1, 2, 3** and **6**

It is a *common* factor when it is a factor of two (or more) numbers.

Let's start with an Example ...

Greatest Common Factor of 12 and 16

- Find all the **Factors** of each number,
- Circle the **Common** factors,
- Choose the **Greatest** of those



What is the "Greatest Common Factor" ?

It is simply the **largest** of the common factors.

In our previous example, the largest of the common factors is 15, so the **Greatest Common Factor** of 15, 30 and 105 is **15**

The "Greatest Common Factor" is the largest of the common factors (of two or more numbers)

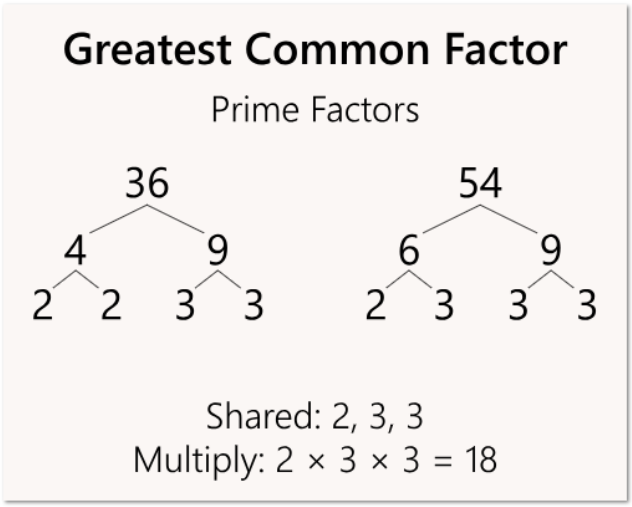
FINDING THE GREATEST COMMON FACTOR

Find the prime factors and combine the common ones together:

Or we can find the prime factors and combine the common ones together:

Two Numbers	Thinking ...	Greatest Common Factor	*Example Simplified Fraction*
24 and 108	**2** × **2** × 2 × **3** = 24, and **2** × **2** × **3** × 3 × 3 = 108	2 × 2 × 3 = **12**	$\frac{24}{108} = \frac{2}{9}$

Prime factorization can also be used to determine the greatest common factor. However, rather than multiplying all the prime factors like we did for the least common multiple, we will multiply only the prime factors that the numbers share. The resulting product is the greatest common factor.



Least Common Multiple

The smallest positive number that is a multiple of two or more numbers.

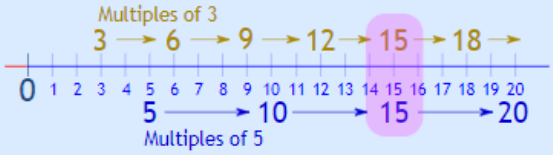
Let's start with an Example ...

Example: Least Common Multiple of 3 and 5:

List the **Multiples** of each number,

The multiples of **3** are 3, 6, 9, 12, 15, 18, ... etc
The multiples of **5** are 5, 10, 15, 20, 25, ... etc

Find the first **Common** (same) value:



The **Least Common Multiple** of 3 and 5 is **15**

(15 is a multiple of both 3 and 5, and is the smallest number like that.)

What is a "Common Multiple" ?

When we list the multiples of (for example) **4** and **5**, the **common multiples** are those that are found in both lists:

The multiples of 4 are: 4,8.12,16,**20**,24.28,32.36,**40**,44,...

The multiples of 5 are: 5.10,15,**20**,25.30,35,**40**,45.50,...

Notice that **20** and **40** appear in both lists?

So, the common multiples of 4 and 5 are: **20, 40,** (and **60**, **80**, etc ..., too)

What is the "Least Common Multiple" ?

It is simply the **smallest** of the common multiples.

In our previous example, the smallest of the common multiples is **20** ...

... so the **Least** Common Multiple of 4 and 5 is **20**.

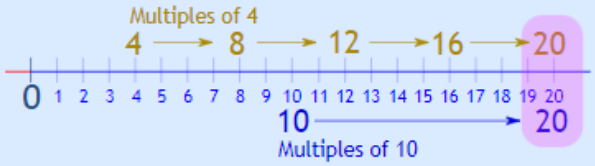
Finding the Least Common Multiple

List the multiples of the numbers until we get our first match.

Example: Find the least common multiple of 4 and 10:

The multiples of 4 are: ***4, 8, 12, 16, 20, ...***
and the multiples of 10 are: ***10, 20, ...***

Aha! there is a match at 20. It looks like this:



So the least common multiple of 4 and 10 is **20**

Once the prime factors are determined, list the shared factors once, and then multiply them by the other remaining prime factors. The result is the least common multiple:

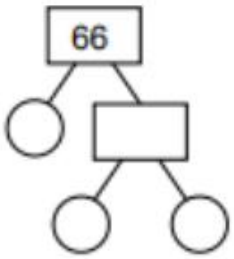
$$36 = 2 \times 2 \times 3 \times 3$$

$$90 = 2 \times 3 \times 3 \times 5$$

$$\text{LCM} = 2 \times 3 \times 3 \times 2 \times 5$$

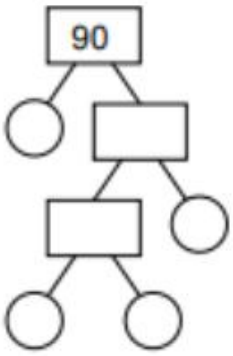
Find the Prime Factors of the Numbers

1)



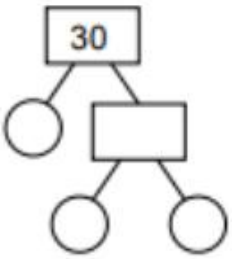
Prime Factors
_ x _ x _ = 66

2)



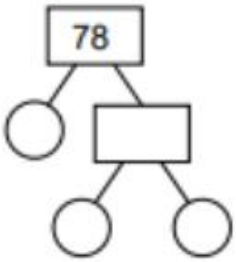
Prime Factors
_ x _ x _ x _ = 90

3)



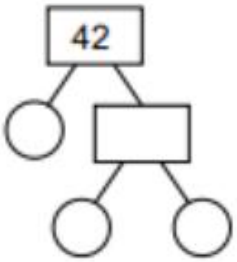
Prime Factors
_ x _ x _ = 30

4)



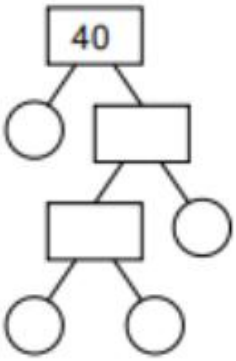
Prime Factors
_ x _ x _ = 78

5)



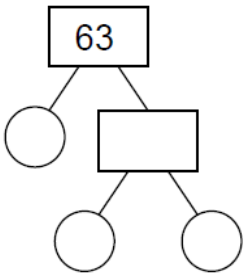
Prime Factors
_ x _ x _ = 42

6)



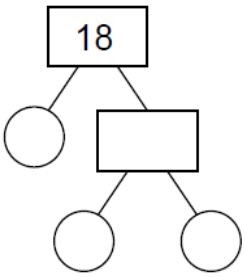
Prime Factors
_ x _ x _ x _ = 40

1)



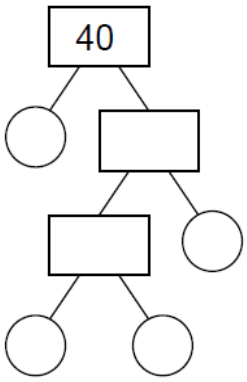
Prime Factors
_ x _ x _ = 63

2)



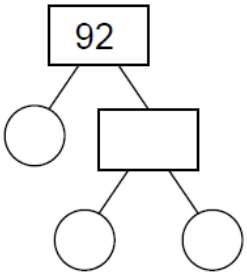
Prime Factors
_ x _ x _ = 18

3)



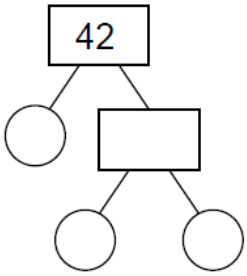
Prime Factors
_ x _ x _ x _ = 40

4)



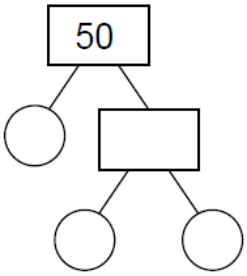
Prime Factors
_ x _ x _ = 92

5)



Prime Factors
_ x _ x _ = 42

6)



Prime Factors
_ x _ x _ = 50

1 Find the LCM of:

a	2 and 5	**b**	8 and 10	**c**	6 and 4
d	3 and 9	**e**	35 and 55	**f**	6 and 11
g	2, 4 and 8	**h**	4, 5 and 6	**i**	6, 8 and 9
j	1, 3 and 7	**k**	4, 5 and 8	**l**	3, 4 and 18

1 Find the HCF of each pair of numbers.

a 3 and 6 **b** 24 and 16 **c** 15 and 40 **d** 42 and 70
e 32 and 36 **f** 26 and 36 **g** 22 and 44 **h** 42 and 48

2 Find the HCF of each group of numbers.

a 3, 9 and 15 **b** 36, 63 and 84 **c** 22, 33 and 121

3 Not including the factor provided, find two numbers less than 20 that have:

a an HCF of 2 **b** an HCF of 6

4 What is the HCF of two different prime numbers? Give a reason for your answer.

Applying your skills

5 Simeon has two lengths of rope. One piece is 72 metres long and the other is 90 metres long. He wants to cut both lengths of rope into the longest pieces of equal length possible. How long should the pieces be?

6 Ms Sanchez has 40 canvases and 100 tubes of paint to give to the students in her art group. What is the largest number of students she can have if she gives each student an equal number of canvasses and an equal number of tubes of paint?

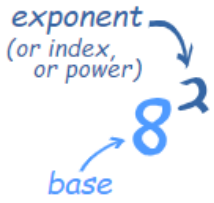
7 Indira has 300 blue beads, 750 red beads and 900 silver beads. She threads these beads to make wire bracelets. Each bracelet must have the same number and colour of beads. What is the maximum number of bracelets she can make with these beads?

EXPONENTS

The **exponent** of a number says **how many times** to use the number in a multiplication.

In 8^2 the "2" says to use 8 twice in a multiplication,

so $8^2 = 8 \times 8 = 64$



In words: 8^2 could be called "8 to the power 2" or "8 to the second power", or simply "8 squared"

What if the Exponent is 1, or 0?

1 If the exponent is 1, then you just have the number itself (example $9^1 = 9$)

0 If the exponent is 0, then you get **1** (example $9^0 = 1$)

Square numbers and square roots

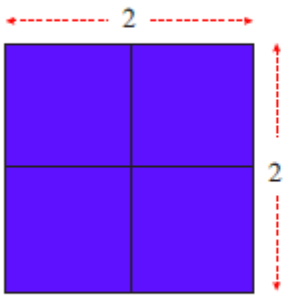
A number is squared when it is multiplied by itself. For example, the square of 5 is $5 \times 5 = 25$. The symbol for squared is ^{2}. So, 5×5 can also be written as 5^2.

The square root of a number is the number that was multiplied by itself to get the square number. The symbol for square root is $\sqrt{}$. You know that $25 = 5^2$, so $\sqrt{25} = 5$.

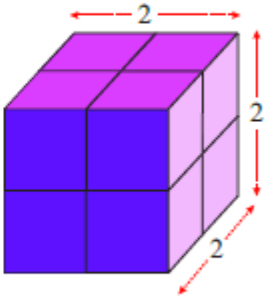
Cube numbers and cube roots

A number is cubed when it is multiplied by itself and then multiplied by itself again. For example, the cube of 2 is $2 \times 2 \times 2 = 8$. The symbol for cubed is ^{3}. So $2 \times 2 \times 2$ can also be written as 2^3.

The cube root of a number is the number that was multiplied by itself to get the cube number. The symbol for cube root is $\sqrt[3]{\ }$. You know that $8 = 2^3$, so $\sqrt[3]{8} = 2$.



a) Square numbers can be arranged to form a square shape. This is 2^2.



b) Cube numbers can be arranged to form a solid cube shape. This is 2^3.

Other powers and roots

You've seen that square numbers are all raised to the power of 2 (5 squared $= 5 \times 5 = 5^2$) and that cube numbers are all raised to the power of 3 (5 cubed $= 5 \times 5 \times 5 = 5^3$). You can raise a number to any power. For example, $5 \times 5 \times 5 \times 5 = 5^4$. This is read as 5 to the power of 4. The same principle applies to finding roots of numbers.

$5^2 = 25$ $\quad\quad$ $\sqrt{25} = 5$

$5^3 = 125$ $\quad\quad$ $\sqrt[3]{125} = 5$

$5^4 = 625$ $\quad\quad$ $\sqrt[4]{625} = 5$

You can use your calculator to perform operations using any roots or squares.

The y^x key calculates any power.

So, to find 7^5, you would enter 7 y^x 5 and get a result of 16 807.

The $\sqrt[x]{\ }$ key calculates any root.

So, to find $4\sqrt{81}$, you would enter 4 $\sqrt[x]{\ }$ 81 and get a result of 3.

LAWS OF EXPONENTS

Law	Example
$x^1 = x$	$6^1 = 6$
$x^0 = 1$	$7^0 = 1$
$x^{-1} = 1/x$	$4^{-1} = 1/4$
$x^m x^n = x^{m+n}$	$x^2x^3 = x^{2+3} = x^5$
$x^m/x^n = x^{m-n}$	$x^6/x^2 = x^{6-2} = x^4$
$(x^m)^n = x^{mn}$	$(x^2)^3 = x^{2\times3} = x^6$
$(xy)^n = x^n y^n$	$(xy)^3 = x^3y^3$
$(x/y)^n = x^n/y^n$	$(x/y)^2 = x^2 / y^2$
$x^{-n} = 1/x^n$	$x^{-3} = 1/x^3$

1 Calculate:

a 3^2 **b** 7^2 **c** 11^2 **d** 12^2 **e** 21^2
f 19^2 **g** 32^2 **h** 100^2 **i** 14^2 **j** 68^2

2 Calculate:

a 1^3 **b** 3^3 **c** 4^3 **d** 6^3 **e** 9^3
f 10^3 **g** 100^3 **h** 18^3 **i** 30^3 **j** 200^3

3 Find a value of x to make each of these statements true.

a $x \times x = 25$ **b** $x \times x \times x = 8$ **c** $x \times x = 121$
d $x \times x \times x = 729$ **e** $x \times x = 324$ **f** $x \times x = 400$
g $x \times x \times x = 8000$ **h** $x \times x = 225$ **i** $x \times x \times x = 1$
j $\sqrt{x} = 9$ **k** $\sqrt{1} = x$ **l** $\sqrt{x} = 81$
m $\sqrt[3]{x} = 2$ **n** $\sqrt[3]{x} = 1$ **o** $\sqrt[3]{64} = x$

4 Use a calculator to find the following roots.

a $\sqrt{9}$ **b** $\sqrt{64}$ **c** $\sqrt{1}$ **d** $\sqrt{4}$ **e** $\sqrt{100}$
f $\sqrt{0}$ **g** $\sqrt{81}$ **h** $\sqrt{400}$ **i** $\sqrt{1296}$ **j** $\sqrt{1764}$
k $\sqrt[3]{8}$ **l** $\sqrt[3]{1}$ **m** $\sqrt[3]{27}$ **n** $\sqrt[3]{64}$ **o** $\sqrt[3]{1000}$
p $\sqrt[3]{216}$ **q** $\sqrt[3]{512}$ **r** $\sqrt[3]{729}$ **s** $\sqrt[3]{1728}$ **t** $\sqrt[3]{5832}$

7 Calculate:

a $(\sqrt{25})^2$	**b** $(\sqrt{49})^2$	**c** $(\sqrt[3]{64})^3$	**d** $(\sqrt[3]{32})^3$
e $\sqrt{9}+\sqrt{16}$	**f** $\sqrt{9+16}$	**g** $\sqrt{36}+\sqrt{64}$	**h** $\sqrt{36+64}$
i $\sqrt{100-36}$	**j** $\sqrt{100}-\sqrt{36}$	**k** $\sqrt{25}\times\sqrt{4}$	**l** $\sqrt{25\times4}$
m $\sqrt{9\times4}$	**n** $\sqrt{9}\times\sqrt{4}$	**o** $\sqrt{\frac{36}{4}}$	**p** $\frac{\sqrt{36}}{4}$

8 Find the length of the edge of a cube with a volume of:

a $1000\,\text{cm}^3$ **b** $19\,683\,\text{cm}^3$ **c** $68\,921\,\text{mm}^3$ **d** $64\,000\,\text{cm}^3$

9 If the symbol * means 'add the square of the first number to the cube of the second number', calculate:

a 2 * 3	**b** 3 * 2	**c** 1 * 4	**d** 4 * 1	**e** 2 * 4
f 4 * 2	**g** 1 * 9	**h** 9 * 1	**i** 5 * 2	**j** 2 * 5

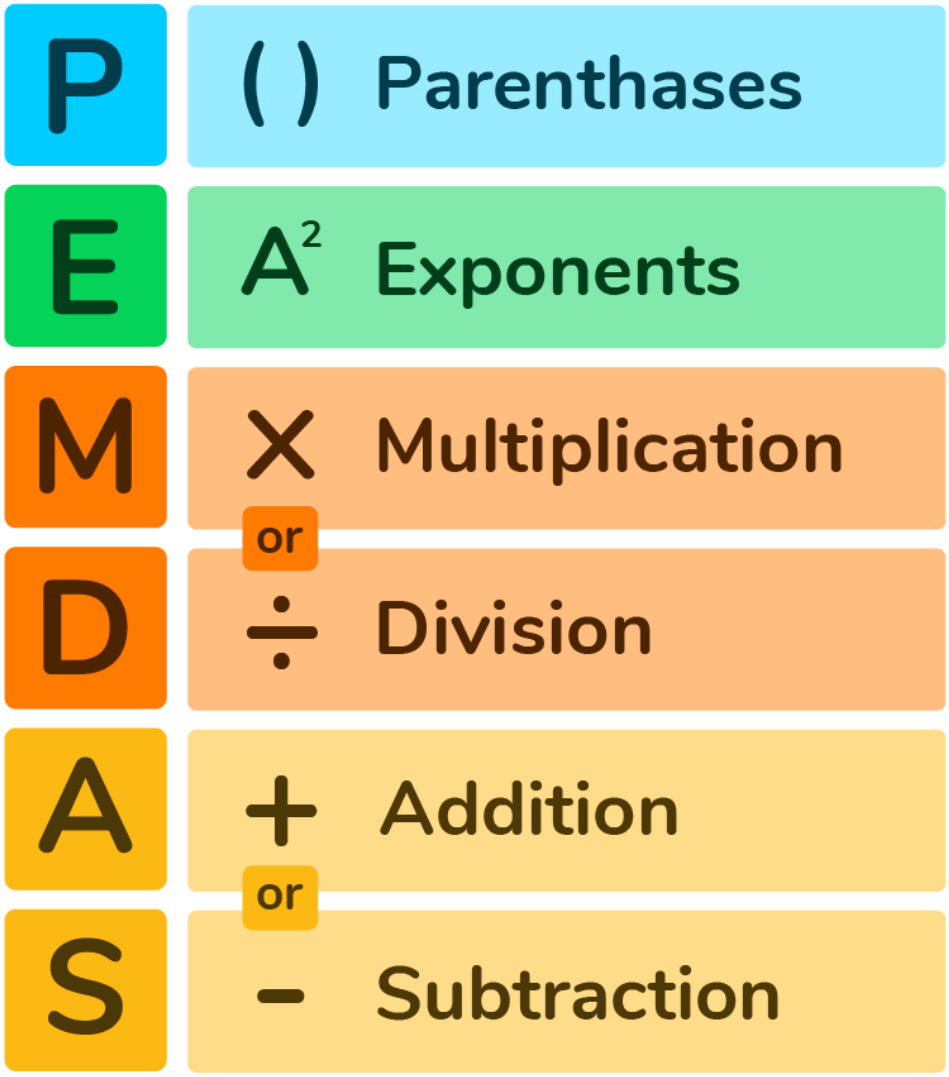
10 Evaluate.

a $2^4\times2^3$	**b** $3^5\times\sqrt[6]{64}$	**c** $3^4+\sqrt[4]{256}$
d $2^4\times\sqrt[5]{7776}$	**e** $\sqrt[4]{625}\times2^6$	**f** $8^4\div(\sqrt[5]{32})^3$

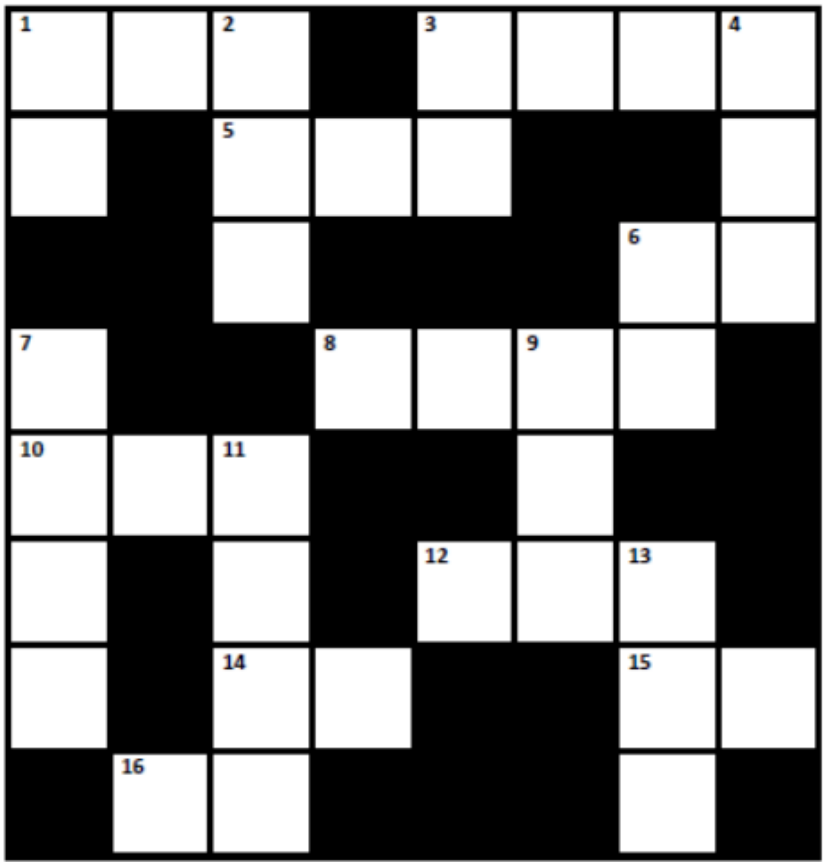
Activity (last page)

$42:4^1$	A:2	$14^1\times14^1$	B:1	$13^{20}:13^{18}$	D:2	$9^8:9^5$	C:1
$85:8^2$	B:2	$7^9:7^6$	C:3	$6^{14}:6^{12}$	A:1	$(16^2)^1$	D:1
$12^6:12^4$	C:4	$3^2\times3^2$	D:3	$5^{12}:5^9$	B:3	$(2^3)^2$	A:3
$2^4\times2^3$	D:4	$(4324)^1$	A:4	$6^{12}:6^9$	C:2	$(4324)^0$	B:4

ORDER OF OPERATIONS



1. Solve the following puzzle



Across

1: 12^2
3: $1001 \times 5 - 4$
5: $5 \times 8 + 8^2$
6: $7 + 8 \times 9$
8: $10^3 + 10$
10: $121 \div 11 \times 11$
12: $90 \times 10 + 11$
14: $99 \div 9 \times 5$
15: $165 \div 15$
16: $8 \times 5 \div 2$

Down

1: $4 + 3 \times 5$
2: $20 \times 10 \times 2 + 12$
3: $9 \times 3 \times 2$
4: $7^2 + 10^2$
6: $8 \times 8 + 6$
7: $3 \times 7 \times 101$
9: 11^2
11: $200 \times 10 - 50$
13: $4 + 3^3 \times 2^2$

EXERCISES

1 Calculate. Show the steps in your working.

a $(4+7)\times 3$	**b** $(20-4)\div 4$	**c** $50\div(20+5)$	**d** $6\times(2+9)$
e $(4+7)\times 4$	**f** $(100-40)\times 3$	**g** $16+(25\div 5)$	**h** $19-(12+2)$
i $40\div(12-4)$	**j** $100\div(4+16)$	**k** $121\div(33\div 3)$	**l** $15\times(15-15)$

2 Calculate:

a $(4+8)\times(16-7)$	**b** $(12-4)\times(6+3)$	**c** $(9+4)-(4+6)$
d $(33+17)\div(10-5)$	**e** $(4\times 2)+(8\times 3)$	**f** $(9\times 7)\div(27-20)$
g $(105-85)\div(16\div 4)$	**h** $(12+13)\div 5^2$	**i** $(56-6^2)\times(4+3)$

3 Simplify. Remember to work from the innermost grouping symbols to the outermost.

a $4+[12-(8-5)]$	**b** $6+[2-(2\times 0)]$
c $8+[60-(2+8)]$	**d** $200-[(4+12)-(6+2)]$
e $200\times\{100-[4\times(2+8)]\}$	**f** $\{6+[5\times(2+30)]\}\times 10$
g $[(30+12)-(7+9)]\times 10$	**h** $6\times[(20\div 4)-(6-3)+2]$
i $1000-[6\times(4+20)-4\times(3+0)]$	

4 Calculate:

a $6+72$	**b** $29-23$	**c** 8×42
d $20-4\div 2$	**e** $\dfrac{31-10}{14-7}$	**f** $\dfrac{100-40}{5\times 4}$
g $\sqrt{100-36}$	**h** $\sqrt{8+8}$	**i** $\sqrt{90-9}$

5 Insert brackets into the following calculations to make them true.

a $3\times 4+6=30$	**b** $25-15\times 9=90$	**c** $40-10\times 3=90$
d $14-9\times 2=10$	**e** $12+3\div 5=3$	**f** $19-9\times 15=150$
g $10+10\div 6-2=5$	**h** $3+8\times 15-9=66$	**i** $9-4\times 7+2=45$
j $10-4\times 5=30$	**k** $6\div 3+3\times 5=5$	**l** $15-6\div 2=12$
m $1+4\times 20\div 5=20$	**n** $8+5-3\times 2=20$	**o** $36\div 3\times 3-3=6$
p $3\times 4-2\div 6=1$	**q** $40\div 4+1=11$	**r** $6+2\times 8+2=24$

1 Simplify. Show the steps in your working.

a $5\times 10+3$	**b** $5\times(10+3)$	**c** $2+10\times 3$
d $(2+10)\times 3$	**e** $23+7\times 2$	**f** $6\times 2\div(3+3)$

g	$\frac{15-5}{2\times5}$	**h**	$(17+1)\div9+2$	**i**	$\frac{16-4}{4-1}$
j	$17+3\times21$	**k**	$48-(2+3)\times2$	**l**	$12\times4-4\times8$
m	$15+30\div3+6$	**n**	$20-6\div3+3$	**o**	$10-4\times2\div2$

2 Simplify:

a	$18-4\times2-3$	**b**	$14-(21\div3)$	**c**	$24\div8\times(6-5)$
d	$42\div6-3-4$	**e**	$5+36\div6-8$	**f**	$(8+3)\times(30\div3)\div11$

3 State whether the following are true or false.

a	$(1+4)\times20+5=1+(4\times20)+5$	**b**	$6\times(4+2)\times3>(6\times4)\div2\times3$
c	$8+(5-3)\times2<8+5-(3\times2)$	**d**	$100+10\div10>(100+10)\div10$

4 Place the given numbers in the correct spaces to make a correct number sentence.

a	0, 2, 5, 10	□−□÷□=□
b	9, 11, 13, 18	□−□÷□=□
c	1, 3, 8, 14, 16	□÷(□−□)−□=□
d	4, 5, 6, 9, 12	(□+□)−(□−□)=□

- Work out each of the following:

a.	4×18÷3=	e.	2+3×4=	i.	10÷5+7=
b.	8+4÷6=	f.	4÷3-3=	j.	12÷4×3=
c.	3·2+3=	g.	4-3×5=	k.	13+3×5×2÷6=
d.	2+3×4=	h.	4+3×5=	l.	1×4+3+8÷2=

1. Work out each of the following:
   a. 3·2+5
   b. 3·7+4·5-12·2
   c. 7+8·17-5-10:2·5
   d. 10-3·2+35÷5+3·2-4

2. Which of the following is the numerical expression for “forty-four subtracted from eighty”?
   a. 44-80
   b. 44+80
   c. 80+44
   d. 80-44

3. Which of the following is the numerical expression for “the quotient of thirty-six and six”?
   a. 36-6
   b. 36x6
   c. 36+6
   d. 36:6

4. Solve: 3×5-7×2+5=

a. 6
b. 2
c. 21
d. Not defined

5. Solve: 3-5×0+2:2=
   a. Not defined
   b. 2
   c. 0
   d. 1
6. 8+1=9

   Which of the following would be the written expression for the operation above?
   a. Eight plus one equals to nine
   b. The sum of eight and one is nine
   c. If we add eight and one, the result is nine
   d. All the answers are correct
7. Multiply two and four, then add six and divide the result by seven and from that result subtract two. The final result is:
8. What is the quotient of twenty-five and five?
9. What is the difference between eighteen and four?
10. If we have number 20, what is the following odd number?
11. The product of three and six is:
12. What is the correct statement?
    a. 25 is a multiple of 6
    b. 8 is a factor of 5
    c. 24 is a multiple of 4
    d. 7 is a multiple of 35

ARITHMETIC WORD PROBLEMS

They test your ability to correctly apply arithmetic operations in a problem situation.
You need to identify:

- What the problem is about.
- Which quantities are being given
- What is being asked for
- Which arithmetic operations must be applied?

The Basic Operations

In basic mathematics there are many ways of saying the same thing:

Symbol	Words Used
+	Addition, Add, Sum, Plus, Increase, Total
−	Subtraction, Subtract, Minus, Less, Difference, Decrease, Take Away, Deduct
×	Multiplication, Multiply, Product, By, Times, Lots Of
÷	Division, Divide, Quotient, Goes Into, How Many Times

EXERCISES

1. There were 46 passengers on the bus. 19 passengers got off at the first stop. 15 passengers got off at the second stop. How many passengers were left on the bus?

2. Linda has $10. She gets $6 a week allowance for doing chores. How many weeks will it take her to save for a video game that costs $52?

3. Alexa is reading a Nancy Drew mystery. There are 95 pages in her book. She read 18 pages the first day. The next day, Alexa just couldn't stop reading. She read 58 pages. How many pages does Alexa have left to read?

4. Allison bought some CDs online. Each CD cost $7. There was an additional charge of $4 per order for shipping costs. The total bill came to $60. How many CDs did Allison buy?

5. Andrew has 5 boards. Each board is 6 feet long. He needs 4 feet of board to make a bird feeder. Andrew wants to make 7 bird feeders. How many feet of board will be left?

6. Amy played checkers with her sister. She won 4 times as many games as she lost. Amy won 12 games. There were no ties. How many games did Amy play?

7. Five friends organized a party. Anna bought 55 cups, Bert the double Anna did. Clara helped with 30 more cups than Bert. Danny was able to buy 65 cups, 25 more than Ethan. How many cups were there in the party?

   a. 410
   b. 370
   c. 405
   d. 395

8. Fred works out 4 times a week for 1 hour each day. During the month of February he didn´t work out for 2 days because he was sick. How many hours did he work out during the whole month?

   a. 16
   b. 8
   c. 14
   d. 10

9. The temperature in Madrid is 22°C. In Buenos Aires there are 10°C less than in Madrid. What is the temperature in Sydney if it is 2 more than in Buenos Aires?

   a. 30
   b. 14
   c. 10
   d. 29

10. Tom needed color pencils for his kids. Each box of 12 pencils was $25. He needed 3 boxes and paid with a $100 bill. How much did he get back?

11. Ms. Smith is making bags of Halloween treats. She wants to put 4 treats in each of the 10 bags she has. If the treats she bought were 43 in total, how many treats are going to be leftover when she fills the ten bags?

12. Melanie went to the store looking for milk. Each bottle contains 1 liter of milk. She bought 8 liters of milk and paid to the cashier $200 and got back $40. How much was each bottle of milk?

13. There are 22 marbles in a bag. 3 of them are green, 5 are red, 4 are yellow and the rest are white. How many white marbles are there in the bag?

14. Joe is having a cook out. He has bought 5 lbs of hamburger for $8.00, and 3 packages of hotdogs for $2.00 each. If Joe paid for all this with a $20 bill, how much money does he have leftover?

    a. $14.00
    b. $6.00
    c. $10.00
    d. He doesn´t have enough money to pay

15. Jack caught 14 fishes on Saturday and twice as much fishes on Sunday. When he went fishing on Monday, he caught half of the fishes he caught on Saturday and Sunday combined. How many fishes did he catch all 3 days?

    a. 63 fishes
    b. 57 fishes
    c. 71 fishes
    d. 59 fishes

16. When asked how old she was, Beth replied “In two years I will be twice as old as I was five years ago”. How old is she?

17. Divide 40 by half and add ten. What is the answer?

18. A farmer has 15 cows, all but 8 died. How many does he have left?

19. There is a pole in a lake. Half of the pole is embedded in the mud at the bottom of the pond, one third of the part that is not embedded in the mud is covered by water, and 6 feet is out of the water. What is the total length of the pole?

20. Orji wants to buy a new bicycle for $1000. He has saved $824. How much more does he need?

21. There are six dogwood trees currently in the park. Park workers had to cut down three dogwood trees that were damaged. How many dogwood trees will be in the park when the workers are finished?

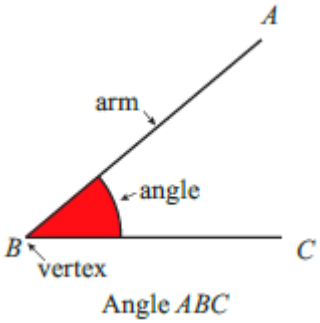
22. Benny has 96 muffins, which he needs to box up into dozens. How many boxes does he need?

23. There are eight rulers in the drawer. Dan placed four more rulers in the drawer. How many rulers are now there in all?

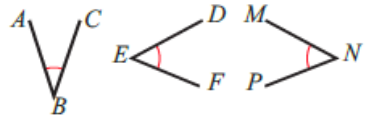
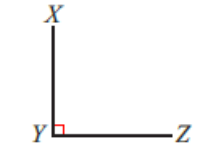
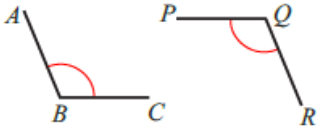
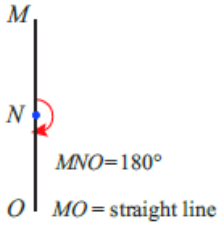
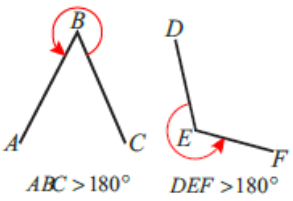
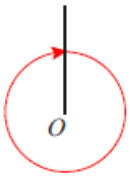
24. There was a total of 18 basketball games in the season. The season is played for 3 months. How many basketball games were played each month, if each month has the same number of games?
25. There are eight hundred ten students at a school. If each classroom holds thirty students, how many classrooms are needed at the school?
26. Mike grew nine cantaloupes, but the rabbits ate seven cantaloupes. How many cantaloupes does Mike have left?
27. Dan found six seashells and Joan found nine seashells on the beach. How many seashells did they find together?
28. Dan bought 6 dozen eggs from the grocery store to bake some cakes. How many eggs did Dan buy?
29. Mike has 3 dozen golf balls. How many golf balls does he have?
30. Tom has two violet marbles. Alyssa has four times more violet marbles than Tom. How many violet marbles do Tom and Alyssa have?

UNIT 2: GEOMETRY AND MEASUREMENTS

ANGLE: When two lines meet at a point, they form an angle. The meeting point is called the vertex of the angle and the two lines are called the arms of the angle. Angles are named using three letters: the letter at the end of one arm, the letter at the vertex and the letter at the end of the other arm. The letter in the middle of an angle name always indicates the vertex.

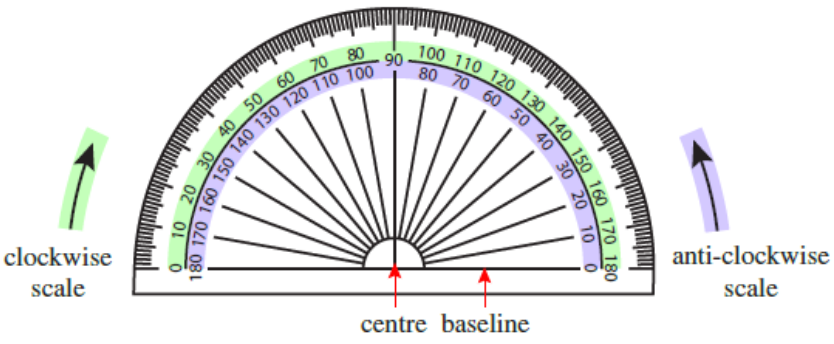


Angle ABC

Acute angle	An acute angle is > 0° but < 90°.	 $ABC < 90°$ $DEF < 90°$ $MNP < 90°$
Right angle	A right angle is an angle of exactly 90°. A square in the corner is usually used to represent 90°. A right angle is formed between perpendicular lines.	 $XYZ = 90°$; $XY \perp YZ$
Obtuse angle	An obtuse angle is > 90° but < 180°.	 $ABC > 90°$ $PQR > 90°$
Straight angle	A straight angle is an angle of 180°. A line is considered to be a straight angle.	 $MNO = 180°$ MO = straight line
Reflex angle	A reflex angle is an angle that is > 180° but < 360°.	 $ABC > 180°$ $DEF > 180°$
Revolution	A revolution is a complete turn; an angle of exactly 360°.	 360°

Measuring and drawing angles

The size of an angle is the amount of turn from one arm of the angle to the other. Angle sizes are measured in degrees (°) from 0 to 360 using a protractor.

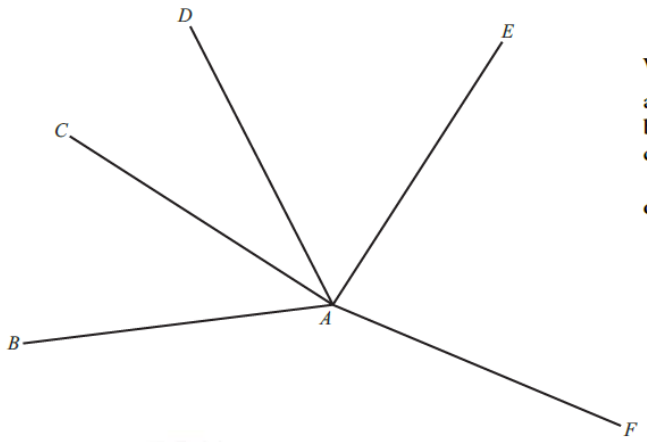


A 180° protractor has two scales. You need to choose the correct one when you measure an angle.

$\therefore ABC = 310°$

Exercise 3.1

1 For each angle listed:



i BAC	ii BAD	iii BAE
iv CAD	v CAF	vi CAE
vii DAB	viii DAE	ix DAF

a state what type of angle it is (acute, right or obtuse)
b estimate its size in degrees
c use a protractor to measure the actual size of each angle to the nearest degree.
d What is the size of reflex angle DAB?

i BAC ii BAD iii BAE
iv CAD v CAF vi CAE
vii DAB viii DAE ix DAF

a state what type of angle it is (acute, right or obtuse)
b estimate its size in degrees
c use a protractor to measure the actual size of each angle to the nearest degree.
d What is the size of reflex angle DAB?

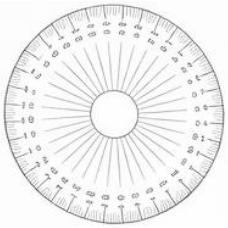
Exercise 3.2

Use a ruler and a protractor to accurately draw the following angles:

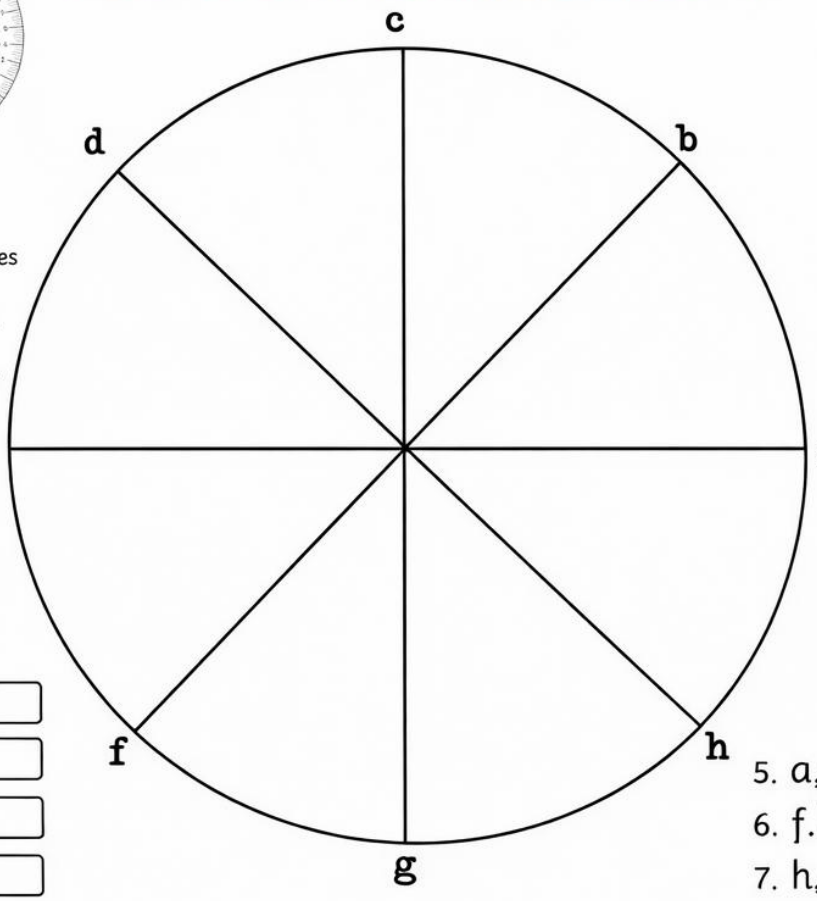
a $ABC = 80°$ b $PQR = 30°$ c $XYZ = 135°$

d $EFG = 90°$ e $KLM = 210°$ f $JKL = 355°$

MEASURING ANGLES 1



Using your protractor, measure the angles of each segment as indicated, e.g. if a" is equal to "a,b."" write your results.



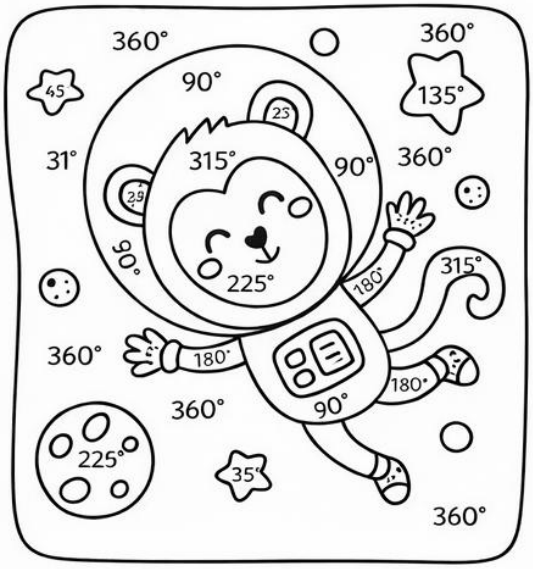
1. Ice Blue
2. Yellow
3. Brown
4. Red

a. a,g= ☐
b. a,d= ☐
c. a,b= ☐
d. a,e= ☐

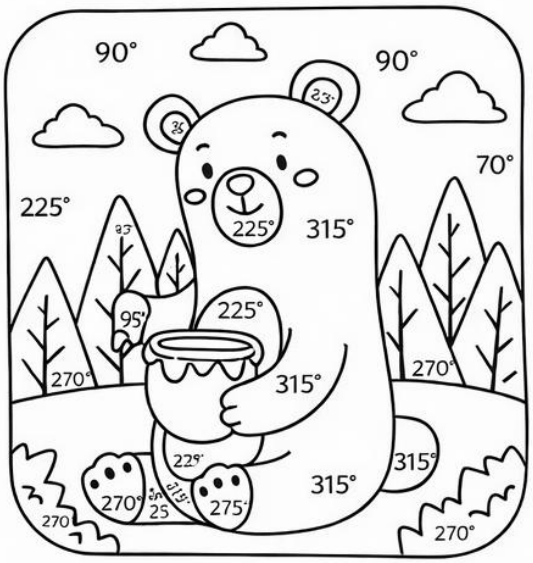
Based on the results drawings according to the color key. Example: if Light Green is result of measuring.ag. it should be colored Light Green.

5. Celeste
6. Brown
7. Green
8. Blue

5. a,c= ☐
6. f.h= ☐
7. h,g= ☐



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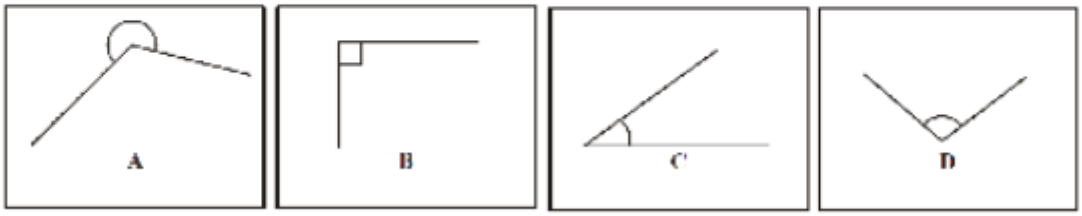


1. Ice Blue _____
2. Yellow _____
3. Red _____

5. Celeste _____
6. Brown _____
7. Green _____

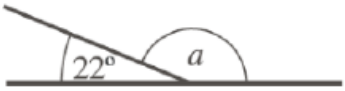
Angles Revision

1. Name the angles below.



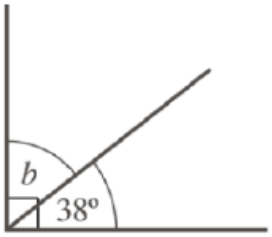
2. Find the missing angles in the questions below.

(a)



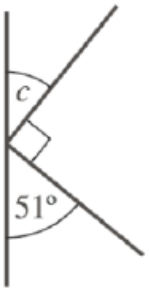
a=

(b)



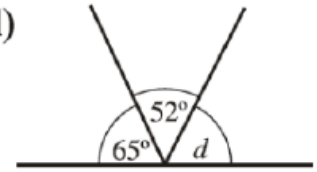
b=

(c)



c=

(d)



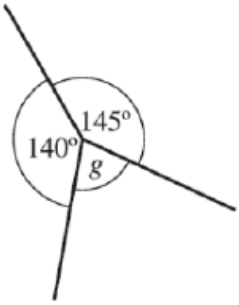
d=

(e)



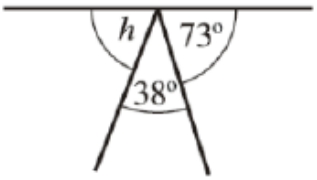
e= f=

(f)



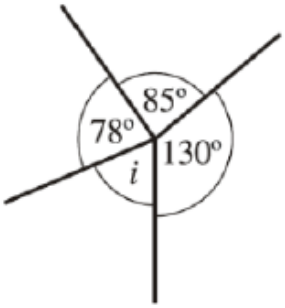
g=

g)



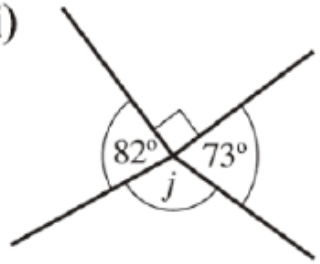
h=

(h)



i=

(i)

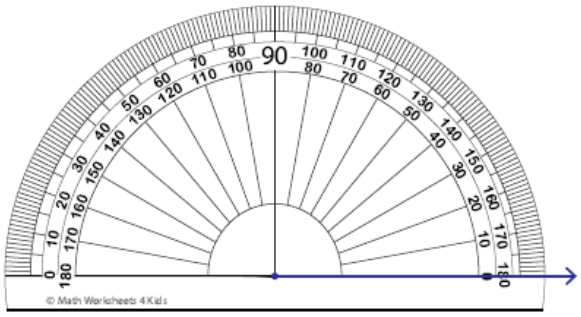


j=

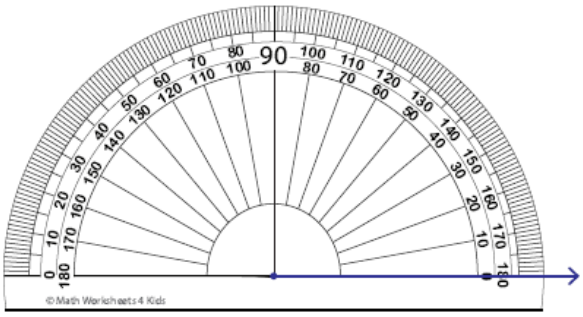
Remember… Angles on a straight line add up to 180°
Angles around a point add up to 360°

Draw each angle using a protractor.

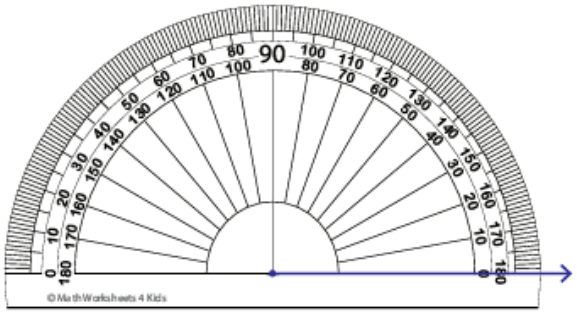
1) 70^0



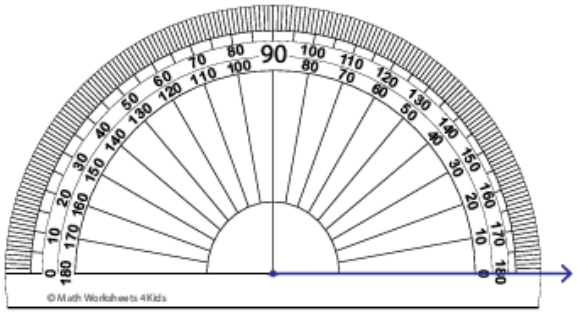
2) 125^0



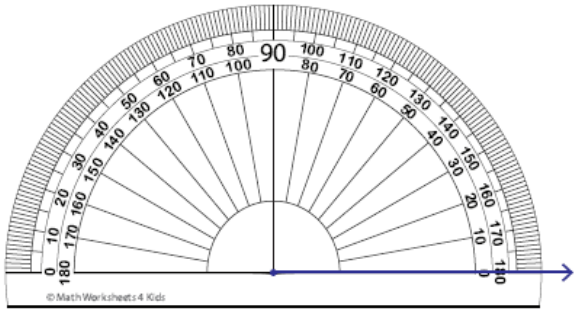
3) 155^0



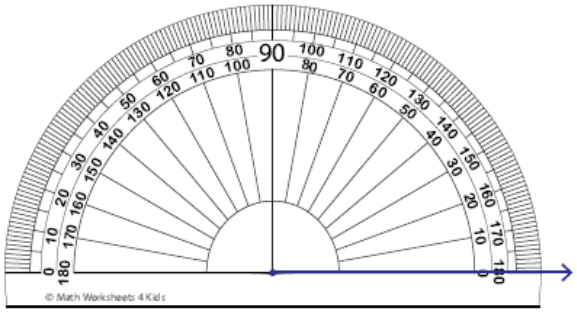
4) 30^0



5) 65^0



6) 140^0



POLYGONS

Polygons are 2-dimensional shapes. They are made of straight lines, and the shape is "closed" (all the lines connect up).

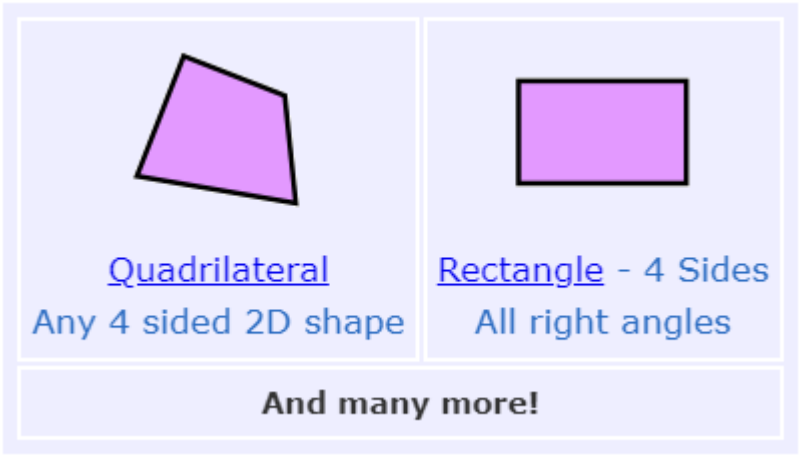
REGULAR POLYGONS

A polygon is a plane (2D) shape with straight sides.

To be a **regular** polygon all the sides and angles must be the same.

NOT REGULAR POLYGONS

These are 2D shapes, and they are polygons, but not regular polygons:



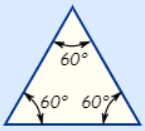
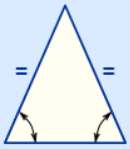
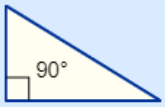
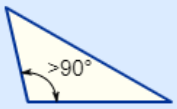
Curved Shapes

These are also 2D shapes, but are **not polygons** because they have curves:



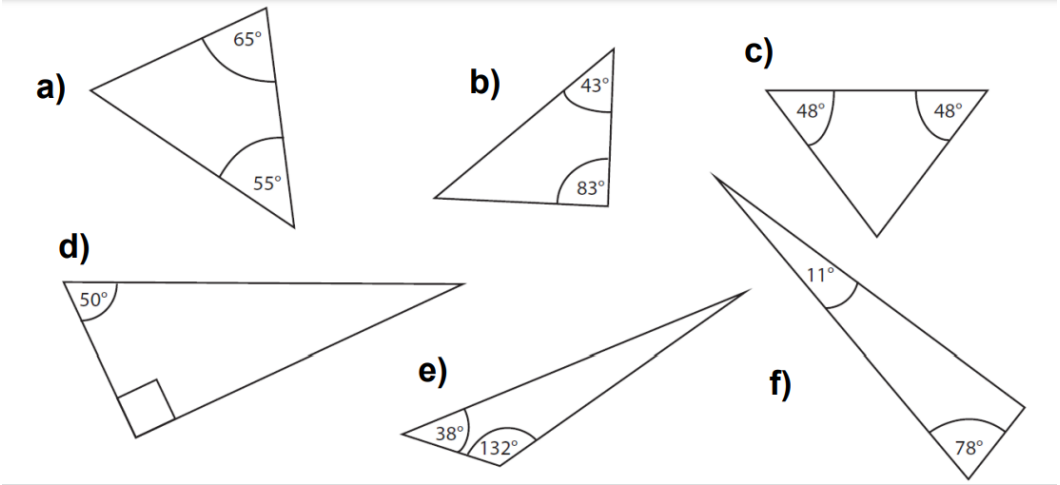
TRIANGLES

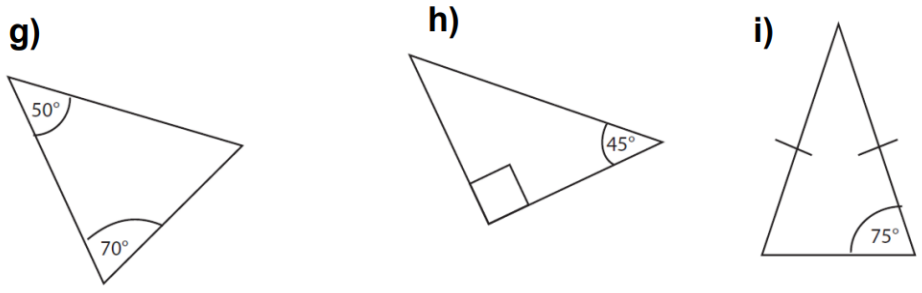
- A triangle has three sides and three angles
- The three angles always add to 180°

BY SIDES	BY ANGLES
 **Equilateral Triangle** **Three** equal sides **Three** equal angles, always 60°	 **Acute Triangle** All angles are less than 90°
 **Isosceles Triangle** **Two** equal sides **Two** equal angles	 **Right Triangle** Has a right angle (90°)
 **Scalene Triangle** **No** equal sides **No** equal angles	 **Obtuse Triangle** Has an angle more than 90°

THE THREE INTERIOR ANGLES ALWAYS ADD TO 180°

- Find the missing angles below





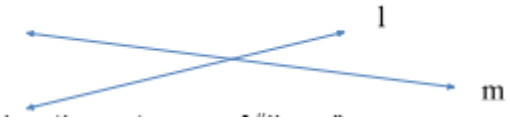
POINTS AND TYPES OF LINES

- **Point:** location. It has no size, no width, no length, and no depth. A point is shown by a dot. They are often labeled with a capital letter.
- **Line**: straight path of points that extends forever in two directions. Traditionally labeled by expressing two points through the line passes. It is written with an arrowhead: $\overleftrightarrow{AB}$.



In this case, points A and B are *collinear*.

Lines may also be labeled with a single scripted letter, and referred to by that name.



We can recognize three types of "lines":

- Line Segment: (or just "segment") part of a line that has defined endpoints, a starting and an ending point. A line segment as the segment between A and B is written as: AB. This is the case of most lines in reality.



- Ray: part of a line that has a starting point but does not end. In the figure above, AB has a starting point A, passes through B and goes forever. It is like "half" of a line.

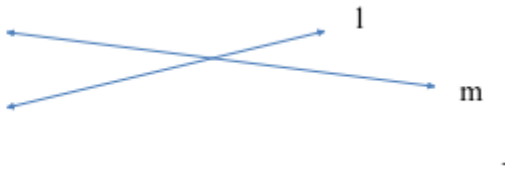


- Line: it has no starting or ending point. We can define $\overleftrightarrow{AB}$ as "the line that passes through the points A and B".

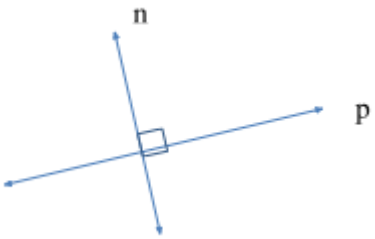


When we work with groups of lines we can find them in different positions. They can touch each other or not. If they never touch, they are *parallel lines*, but if they do, they are *intersecting lines*, they meet in a point.

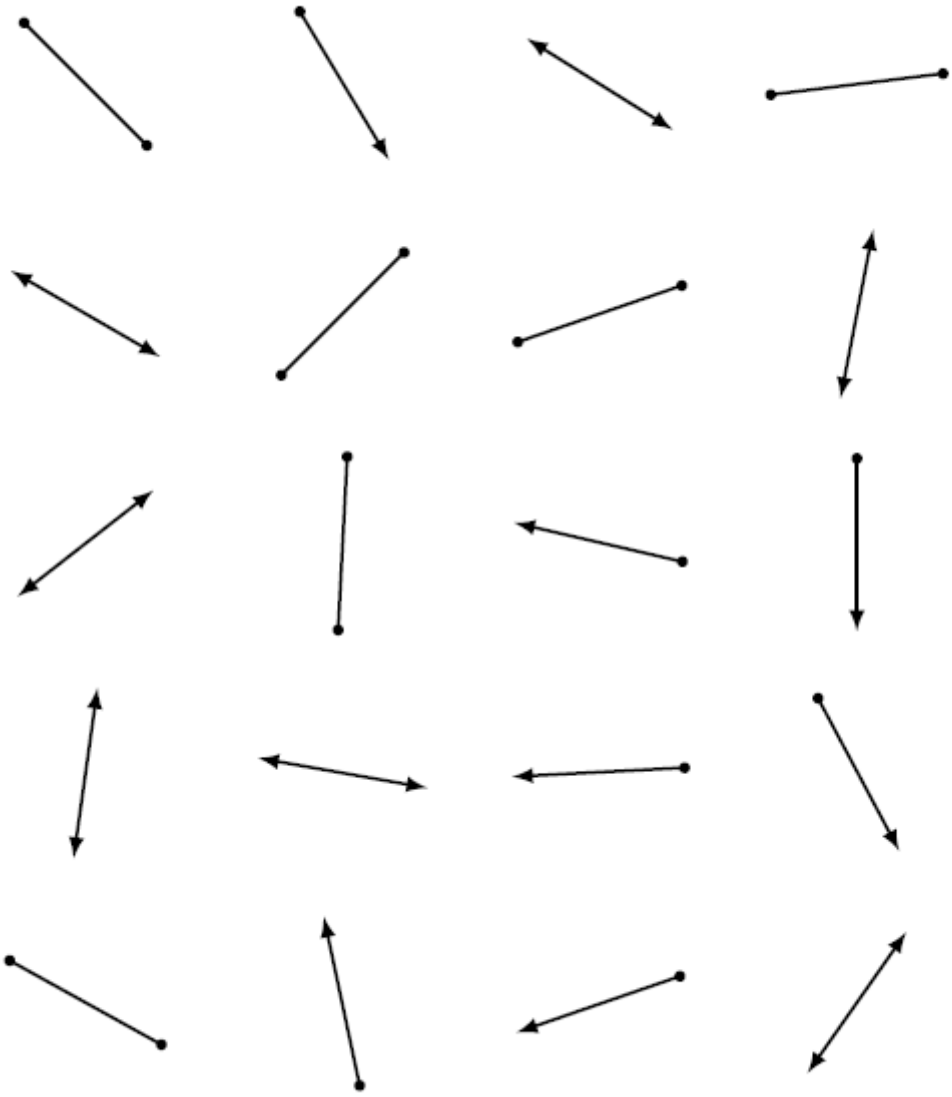




If two lines intersect at a 90-degree angle they are *perpendicular lines.*



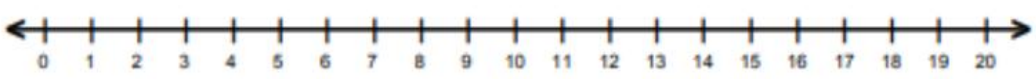
Identify each diagram as representing a line, line segment or a ray.



NUMBER LINES AND SCALES

Lines are infinite. We can draw part of a line and set a scale in order to put numbers in the correct order and make it easier to understand their value.

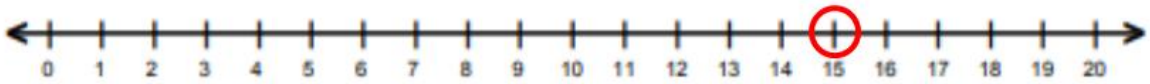
THE DISTANCE BETWEEN TICK MARKS MUST ALWAYS BE THE SAME



The numbers are often shown as specially-marked points evenly spaced on the line. Although this image only shows the numbers from 0 to 20, the line includes all real numbers, continuing forever in each direction, and also numbers not marked that are between the integers.

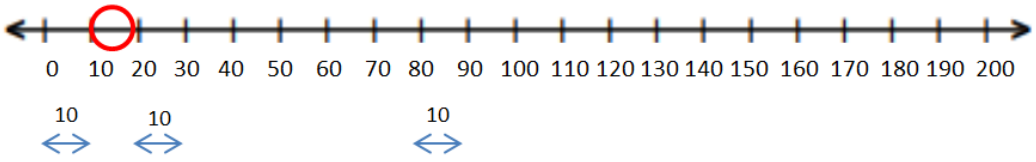
Most of the times, the number we are looking for will not appear on the line, but we can find it by the position it takes and the SCALE that the number line has.

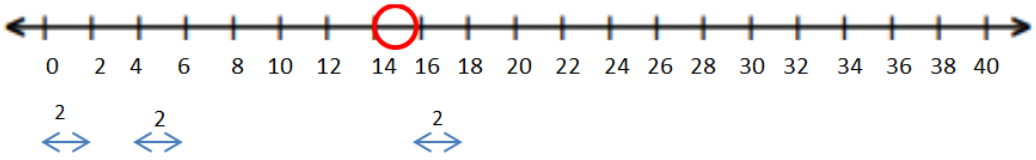
For example: we need to find number 15:



If the scale (value of the distance between tick-marks) changes, then number 15 will be in the middle between 10 and 20 or somewhere between 10 and 12… or somewhere else, depending on the scale we are using.

We can see that the size of the line is the same in both cases, but the value of the distance between tick-marks changes,

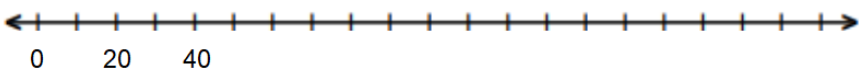


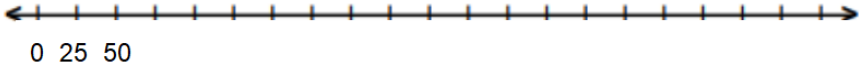


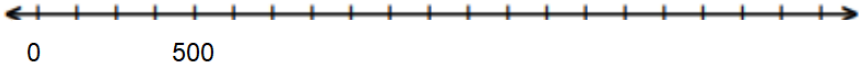
EXERCISES:

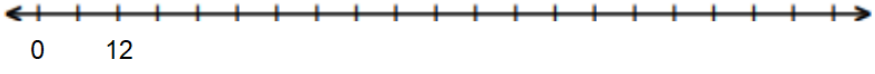
1. Complete the number line according to the scale given.

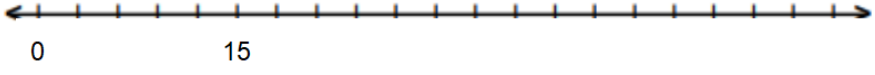
2. Find number 80 on every line. What happens?

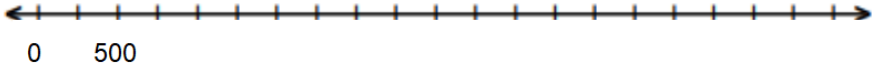










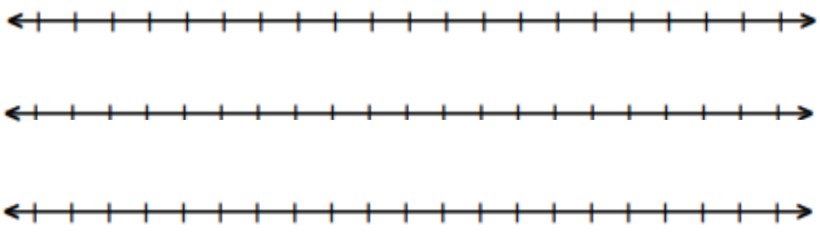


3. Complete the number line with the most suitable scale in order to find clearly the numbers given on each case:

a. 5, 20, 32, 45, 49

b. 120, 540, 600, 830

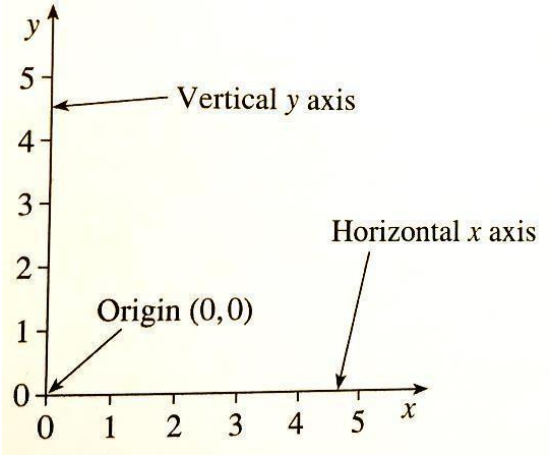
c. 11, 22, 50, 60



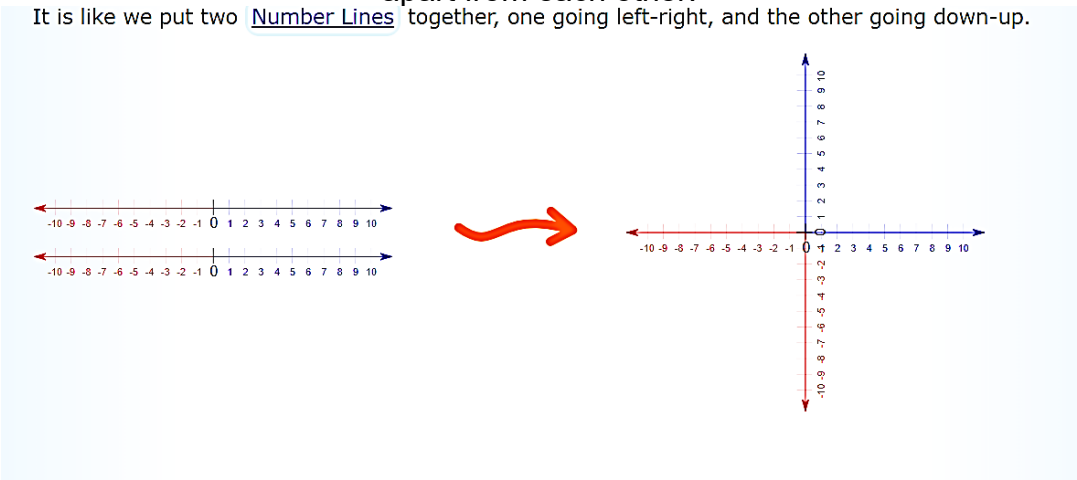
If two lines intersect at a 90-degree angle they are *perpendicular lines.*

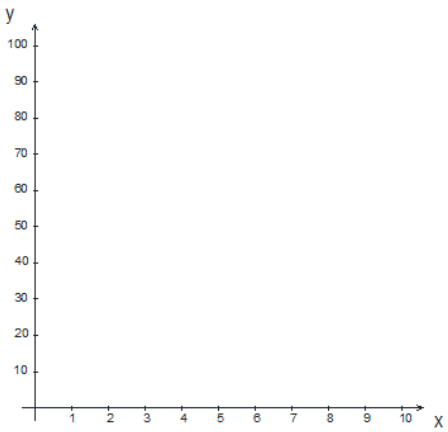
UNIT 3: ALGREBRA AND FUCTIONS

A ***two-dimensional coordinate system*** is formed by two mutually perpendicular lines called *axes*. The axes intersect at the point *O*, which is called the ***origin***. In the right-handed system, one of the axes (x-axis) is directed to the right, the other y-axis is directed vertically upwards.



It is like we put two number lines put together, one going left-right, and the other going down-up. Remember that the tick marks in one number line must be equally apart from each other.

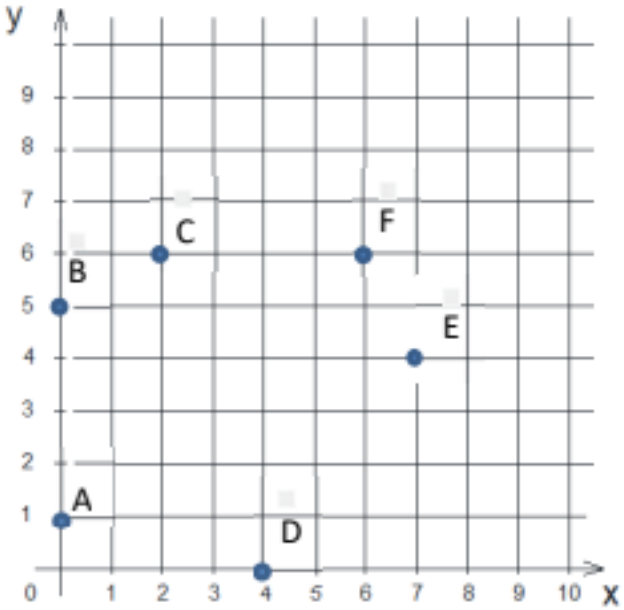




We can find axis which **scales** are different.

LOCATIONS:

The coordinates of any point on the xy-plane are determined by two real numbers x and y, which are orthogonal projections of the points on the respective axes. The x-coordinate of the point is called the abscissa of the point, and the y-coordinate is called its ordinate.
To describe the position or ***location*** of a point in relation to the origin you need coordinates (x,y). The first number tells you how far to move along in a horizontal direction. The second number tells you how far to move up or down in a vertical direction.
The origin has coordinates (0,0). It is where the axes intersect.
A positive *x-coordinate* means move right.
A positive *y-coordinate* means move up.
For example, the coordinates for the locations below are:



. (0,1)

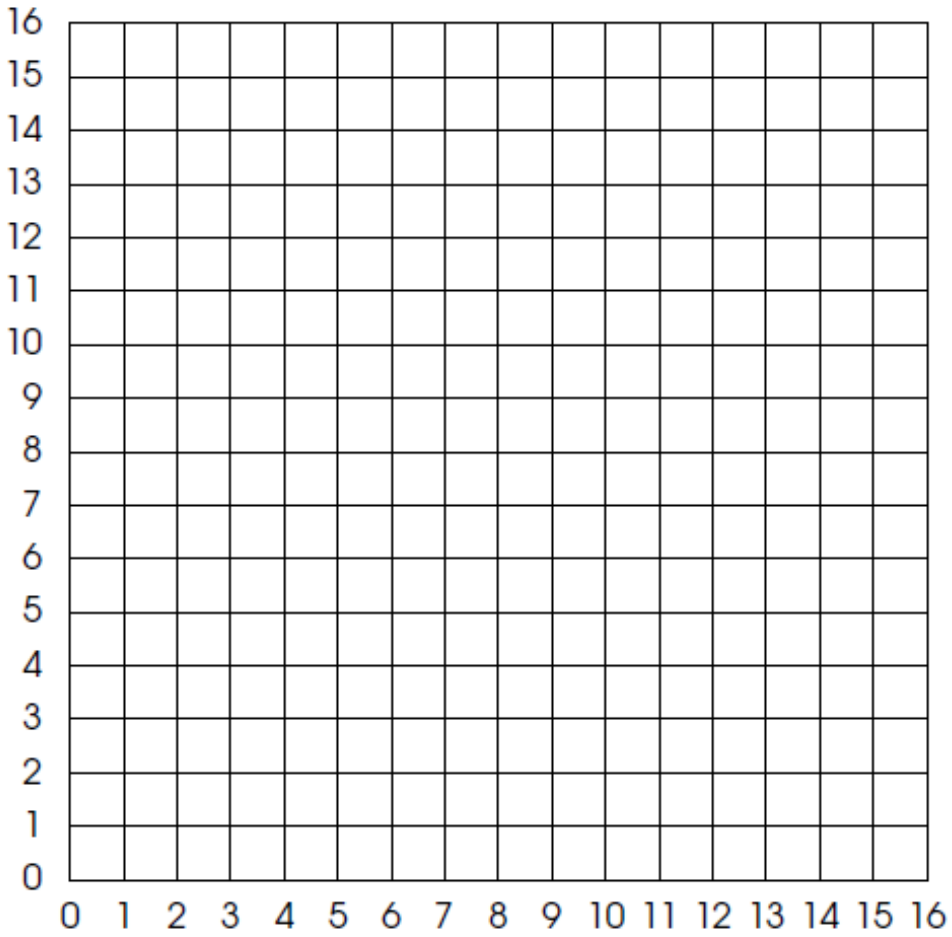
. (0,5)
. (2,6)
. (4,0)
. (7,4)
. (6,6)

- Plot these points on a two-dimensional plane:

(2, 2), (5, 2), (5, 4), (7, 4), (7, 7), (5, 7), (5, 9), (2, 9), (2, 7), (0, 7), (0, 4), (2, 4)

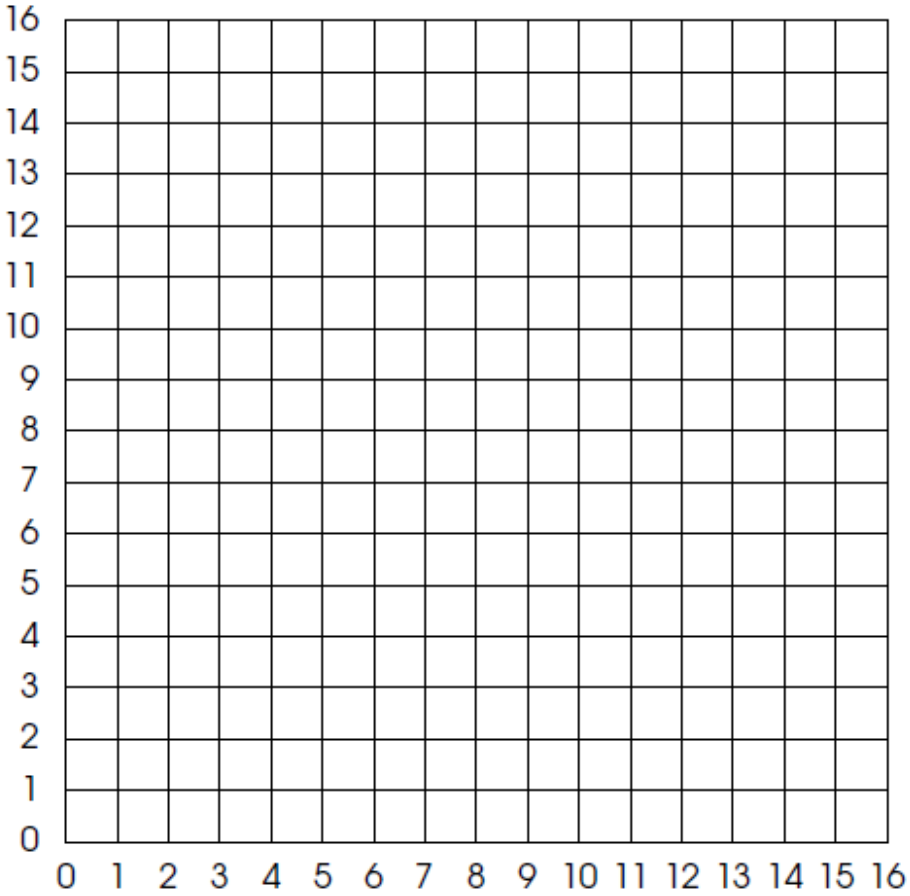
- Connect the points with lines.
- Finally, calculate the area of the shape and its perimeter.

Plot the points and connect them to reveal the drawing.



(8,1) (1,8) (1,11) (3,13) (6,13) (8,11) (10,13) (13,13) (15,11) (15,8) (8,1)

Plot the points and connect them to reveal the drawing.



(0,7) (0,9) (7,16) (8,16) (15,9) (15,7) (13,7) (8,13) (7,13) (2,7) (0,7) ✂ (1,7) (1,0) (14,0) (14,7) ✂ (5,0) (5,6) (10,6) (10,0) ✂ (11,13) (11,15) (13,15) (13,11)

ONLINE GAME: https://www.cokitos.com/ejercicio-de-coordenadas-cartesianas-online/play/

EXERCISES

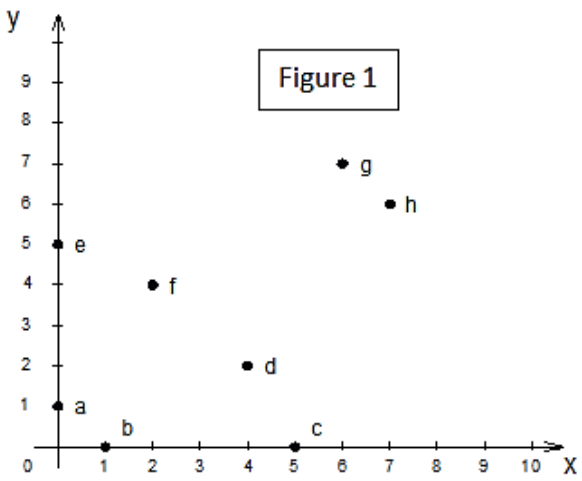
1. Draw a two dimensional system in which you can plot the following locations with a proper scale: (0,5), (10,8), (50,6), (30,3).

2. Draw a two dimensional system in which you can plot the following locations with a proper scale: (5,0), (8,10), (6,50), (3,30).
3. Draw a two dimensional system in which you can plot the following locations with a proper scale: (500,0), (800,10), (600,50), (300,30).

4. Which of the following shows the coordinates for points “a” and “d” in *“Figure 1”*?

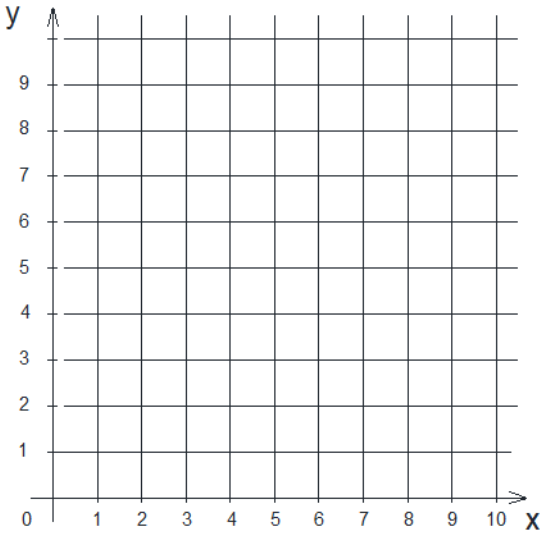
   a) (0,1) and (2,4)
   b) (0,1) and (4,2)
   c) (1,0) and (2,4)
   d) (1,0) and (4,2)

   .



5. Which are the locations of points “b”, “c”, e”, “f”, “g” and “h” on “*Figure 1*”?

6. Place the following points in the two dimensional coordinate system.
   a) (0,6)
   b) (5,1)
   c) (3,0)
   d) (7,2)
   e) (1,8)
   f) (5,0)
   g) (6,4)
   h) (0,2)
   i) (2,0)
   j) (7,7)



7. Draw axes on graphed paper. On the *x-axis*, show the numbers: {1,2,3,4,5,6}. On the *y-axis*, show the numbers: {1,2,…,7,8}.

   Mark these points on the graph: (3, 8), (3, 7), (5,5), (5,4), (4,4), (4,1), (3,1), (3,5), (2,5), (2,1), (1,1), (1,6), (2,6), (1,6), (1, 3), (0,1), (1,0), (6,0), (6,6), (5,7)

   Join each point to the next one in the list. What shape have you made?

Battleship has been a popular game for generations. The original pen-and-paper game has inspired multiple board games, handheld electronic versions, computer games, and even a film. Even after all those versions and rule changes however, the game is still simple enough to be played with graph paper and pens.

How to play:

Hide the chart labeled "my ships" from the other player, and draw a thick outline of the ships anywhere within its boundaries. Each ship is one square wide, and different ranges in length. Follow these rules to determine where to put your ships:

- Ships can be placed horizontally or vertically, but *not* diagonally.
- You must place all the ships on the grid.
- Every ship must be completely on the grid. No ship can hang off the edge of the board.
- Ships cannot overlap each other.
- Once your ships are placed and the game has begun, you are *not* allowed to move your ships again.

Learn how to take a shot. Each player uses the “Enemy ships” grid, without any ships placed on it, to keep track of his/her "shots" at the enemy ships. To take a shot, pick a square on this grid by naming it using the coordinates with the numbers on the left and numbers above the grid.
For example, the square in the top left corner of the grid is named "2-1," since it is in the column labeled 2 and the row labeled 1. (First look at the horizontal numbers, then at the vertical ones).

Learn how to respond to an enemy shot. After player 1 announces where she will be "shooting," player 2 checks the same coordinate square on his lower grid, the one with his ships. Player 2 then responds (telling the truth!) one of the following ways:

- If player 1 hit an empty square, without ships, player 2 says "Miss!"
- If player 1 hit a square with a ship in it, player 2 says "Hit!"

Keep track of shots as they hit or miss. Draw successful hits with Xs and misses with dots, or use any system of symbols you find easy to understand.

Take turns shooting until one player has lost all their ships. Players alternate taking one shot at a time, whether or not that shot is successful. Whoever manages to sink all of her opponent's ships first wins the game.

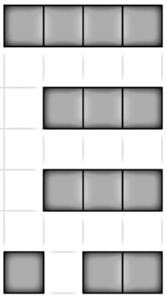
MATH 1st Year 2026

My ships

	1	2	3	4	5	6	7	8	9	10
1										
2										
3										
4										
5										
6										
7										
8										
9										
10										

Enemy ships

	1	2	3	4	5	6	7	8	9	10
1										
2										
3										
4										
5										
6										
7										
8										
9										
10										



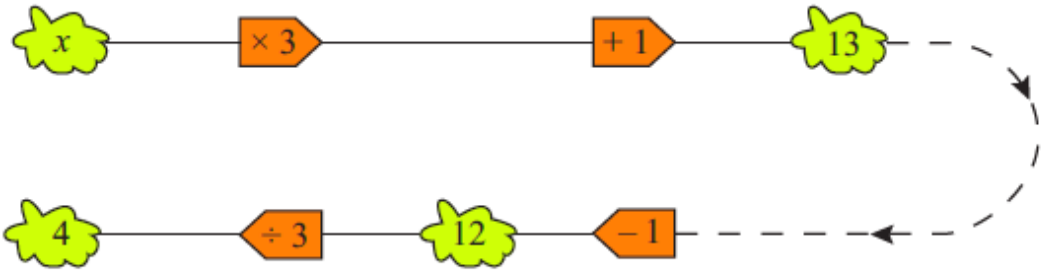
1. FUNCTIONS

A function can be thought of as a rule or formula that tells how to associate the elements in one set (the *domain*) with the elements in another set (the *range*).

I think of a number. My number is x. If I multiply my number by three and then add one, the answer is 13. What is my number?

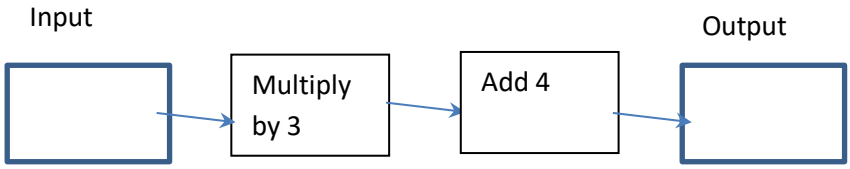
To solve this problem you first need to understand the stages of what is happening to x and then undo them in reverse order:

This diagram (sometimes called a function machine) shows what is happening to x, with the reverse process written underneath. Notice how the answer to the problem appears quite easily:



So $x = 4$

A function machine:

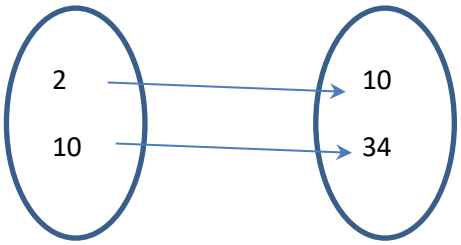


For example:

If 7 is input, then $7 \times 3 + 4 = 25$ is output.

A number enters into the function machine is called the **input**. When a function machine uses a function (in this case *multiply by 3 then add 4*) it produces an **output**. Here, if the input is 2 then the output is 10 because $2 \times 3 + 4 = 10$. If the input is 10 the output is 34, and so on. We can use a mapping diagram like this one to show what outputs go with particular inputs.

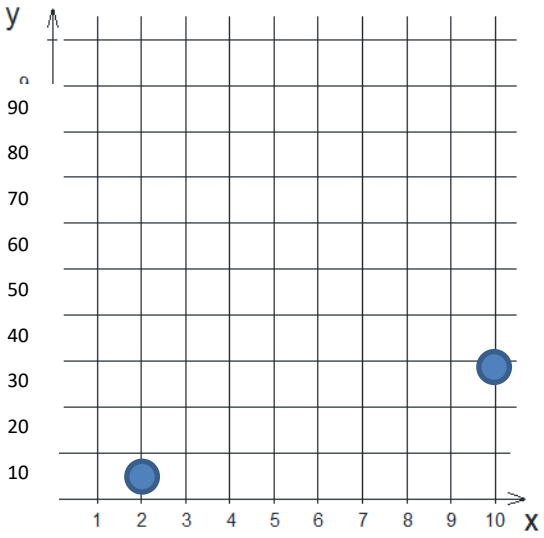
function



Or it can be shown in a table:

Input	Output
2	10
10	34

If we graph the input and output as locations we can see that the input is x and the output is y. then, we will have two locations: (2,10) and (10,34).



The set of numbers in the input: $\{2,10\}$ is what in Math is called the DOMAIN. The set of numbers in the output: $\{10,34\}$ is the RANGE.

Then, the Domain will be the set of all the values for ***x*** and the range the set of all the values of ***y***.

A more compact and efficient solution can be obtained using algebra. Follow the instructions in the question:

1	The number is x:	x
2	Multiply this number by three:	$3x$
3	Then add one:	$3x+1$
4	The answer is 13:	$3x+1=13$

This is called a **linear equation**. 'Linear' refers to the fact that there are no powers of x other than one.

The next point you must learn is that you can change this equation without changing the **solution** (the value of x for which the equation is true) *provided* you do the same to both sides at the same time.

Follow the reverse process shown in the function machine above but carry out the instruction on both sides of the equation:

$$3x + 1 = 13$$

$$3x + 1 - 1 = 13 - 1 \qquad \text{(Subtract one from both sides.)}$$

$$3x = 12$$

$$\frac{3x}{3} = \frac{12}{3} \qquad \text{(Divide both sides by three.)}$$

$$x = 4$$

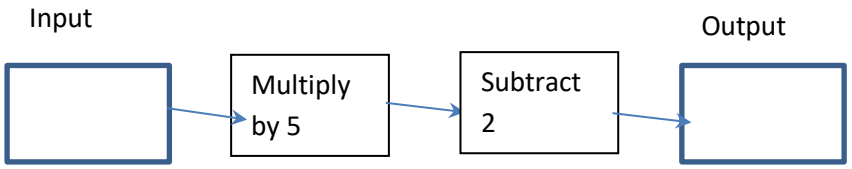
Always line up your '=' signs because this makes your working much clearer.

Sometimes you will also find that linear equations contain brackets, and they can also contain unknown values (like x, though you can use any letter or symbol) on *both* sides.

The following worked example demonstrates a number of possible types of equation.

EXERCISES

1. Complete the table for the function machine:



Input	Output
1	
2	
3	
4	
10	
100	

2. What are the numbers included in the domain? Write them as a set of numbers.

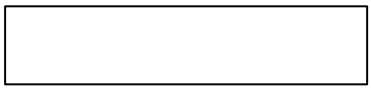
3. What are the numbers included in the range? Write them as a set of numbers.

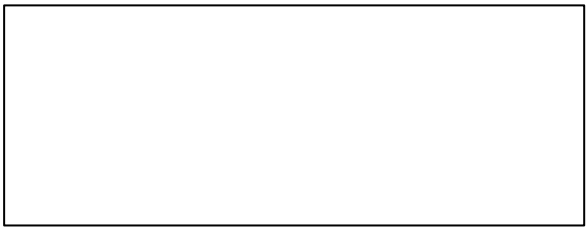
4. Given $y = 3x + 2$, what is the value of y if $x = 2$?
   a) $y = 6$
   b) $y = 4$
   c) $y = 8$

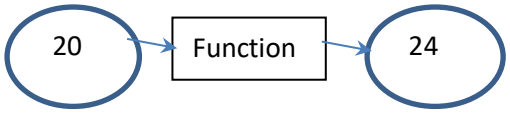
d) $y = 5$

5. In the following function, the inputs are part of the data. Which of the options would be the set of outputs?

x	*y*
0	
1	
2	
5	
3	

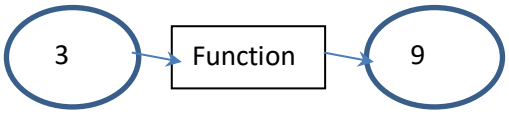




6. 

Given the two numbers in the circles, what of the following would be a possible function?

a) y= x+4
b) y= x+3
c) y= x-4
d) y= x-3

7. 

Given the two numbers in the circles, what of the following would be a possible function?

a) $y = 3 + 3x$
b) $y = 2 + 3x$
c) $y = 1 + 2x$
d) $y = 3 + 2x$

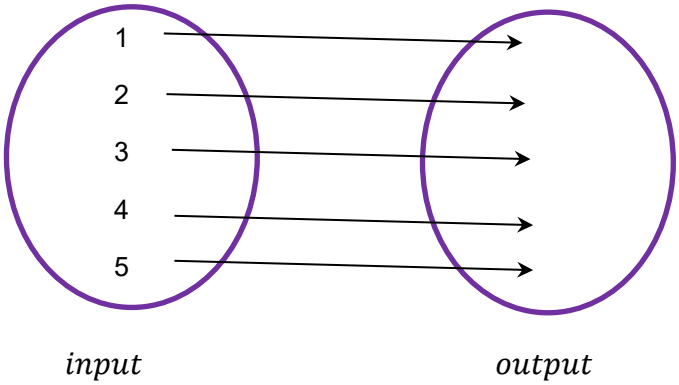
8. Solve if $x = 2$

$$y = 35 - 4x + x : 2$$

9. Copy and complete the mapping diagram using the function shown.

$$function$$

$$x10 + 3$$

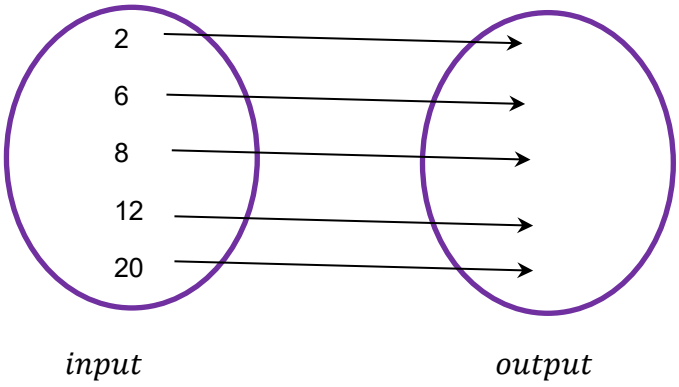


10. Copy and complete these mapping diagrams using the functions shown.

a.

$$function$$

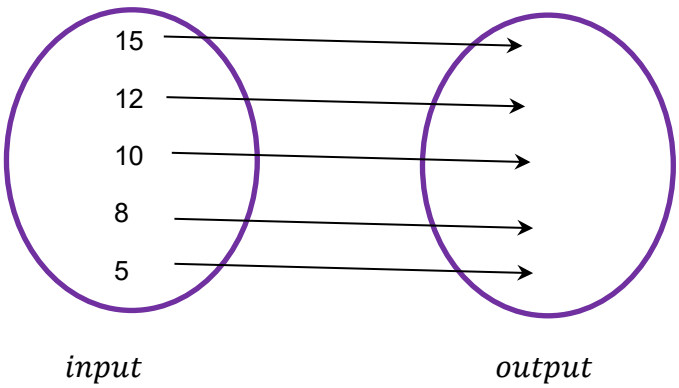
$$\div 2 + 10$$



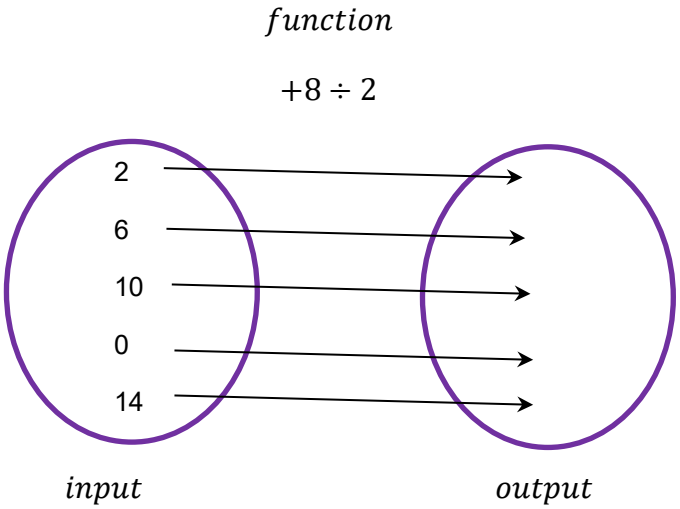
b.

$$function$$

$$-5 \times 3$$



c.



11. Write, as sets, the domain and range of each of the functions on the previous exercise.

a. Domain:

Rage:

b. Domain:

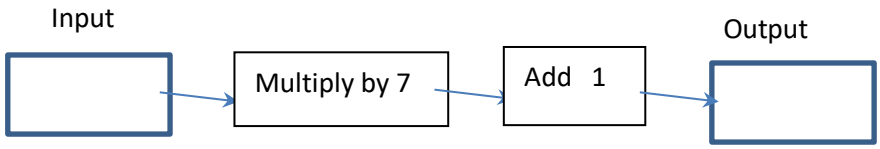
Range:

c. Domain:

Range:

12. Use the function machine to fill in the table:

a.

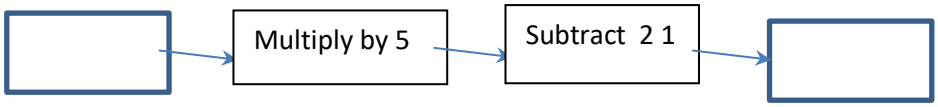


Input	Output
1	
2	
3	
4	
5	

Input

Output

b.

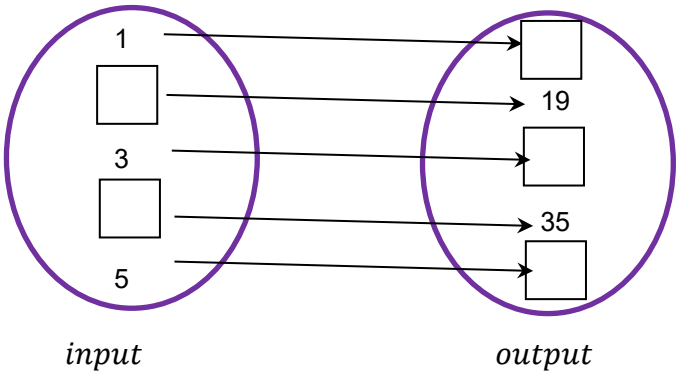


Input	Output
1	
2	
3	
4	
5	

13. Copy and complete these mapping diagrams using the functions shown:

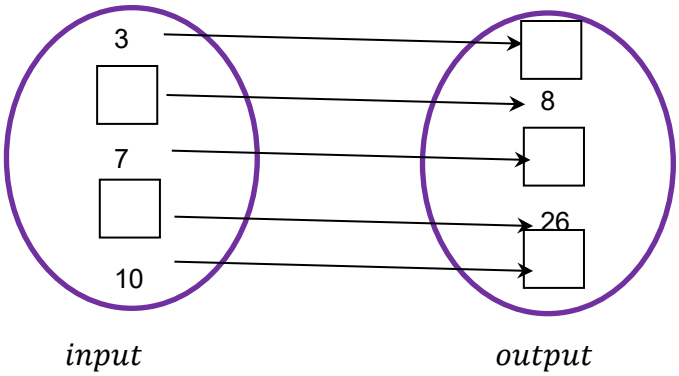
a. $function$

$x8 + 3$



b. $function$

$x6 - 4$

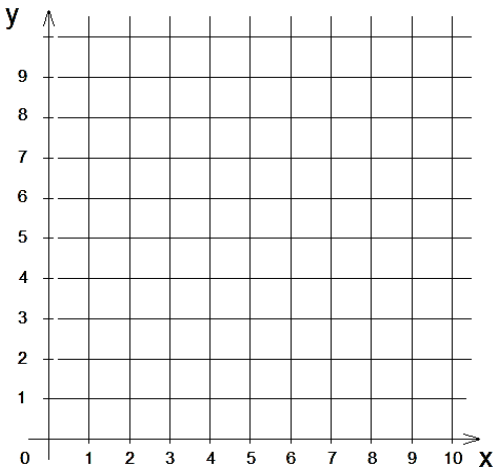


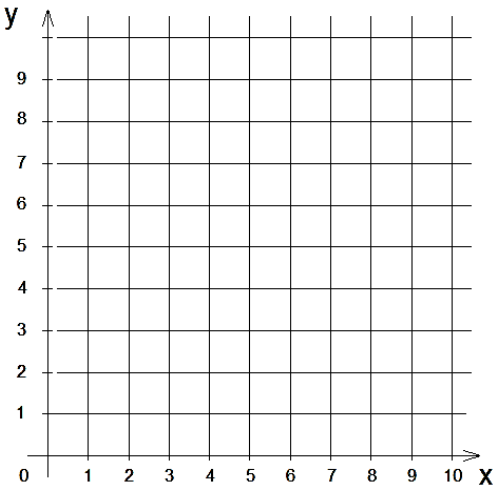
14. Copy and complete the table for each function:
    a. $y = 2x + 2$
    b. $y = x + 5$
    c. $y = x$
    d. $y = 5x$
    e. $y = 10 - x$
    f. $y = (x + 3) \times 4$

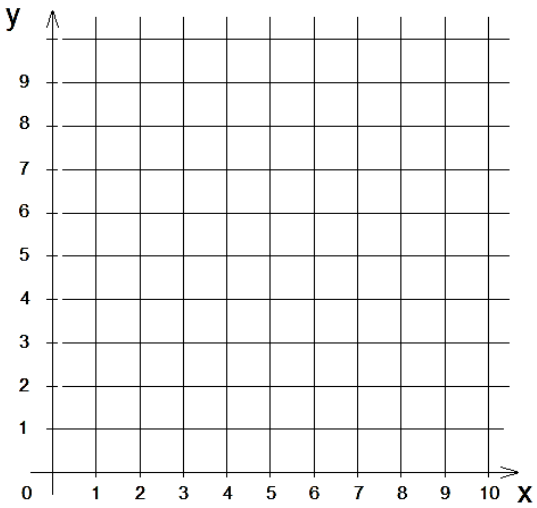
Input	**Output**					
	a	b	c	d	e	f
1						
2						
3						
4						
5						

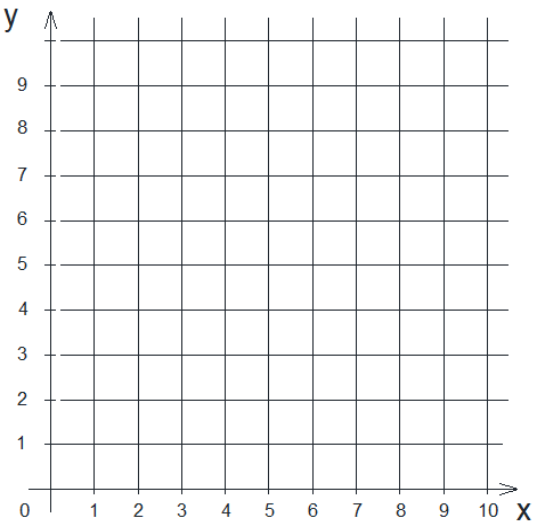
15. Write, as sets, the domain and range of each of the functions on the previous exercise.
    a. Domain:
       Rage:
    b. Domain:
       Range:
    c. Domain:
       Range:
    d. Domain:
       Rage:
    e. Domain:
       Range:
    f. Domain:
       Range:

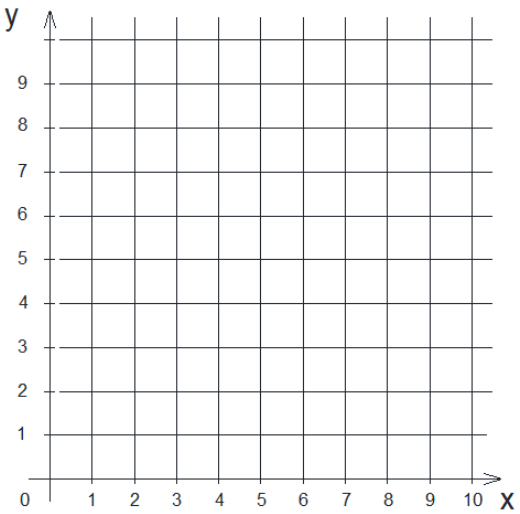
16. Graph the locations given in the previous exercises. Change the scale or add tick marks if necessary.

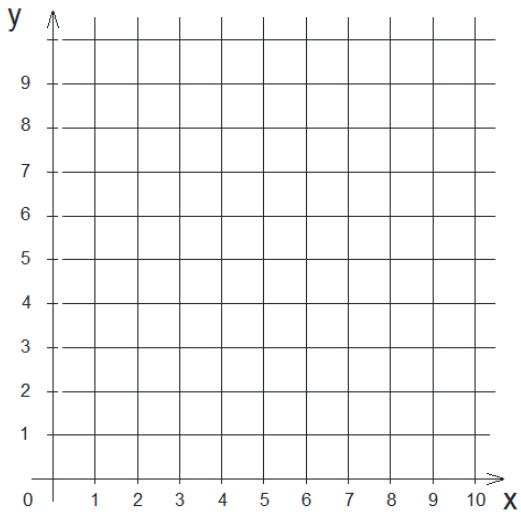

y
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1
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1 2 3 4 5 6 7 8 9 10
X

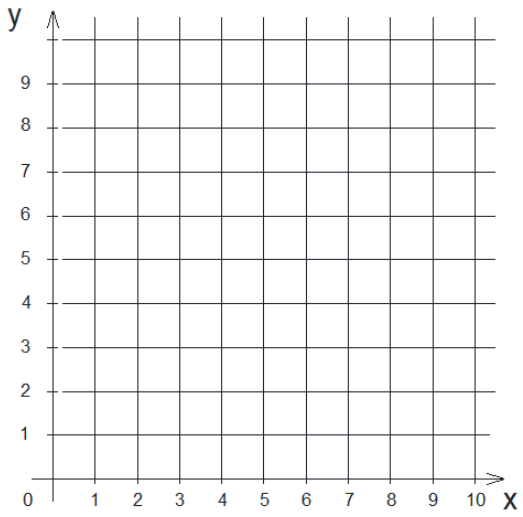

y
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0
1 2 3 4 5 6 7 8 9 10
X

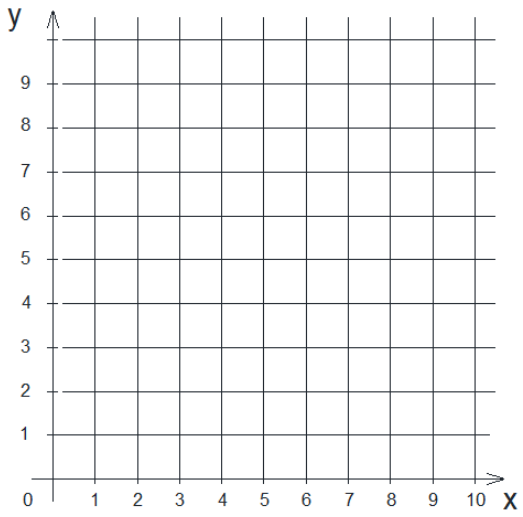

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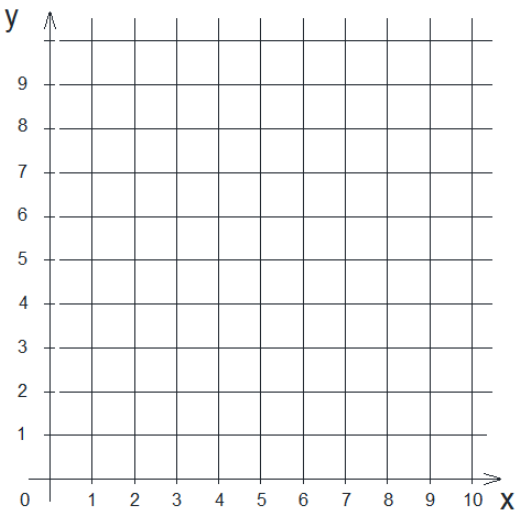

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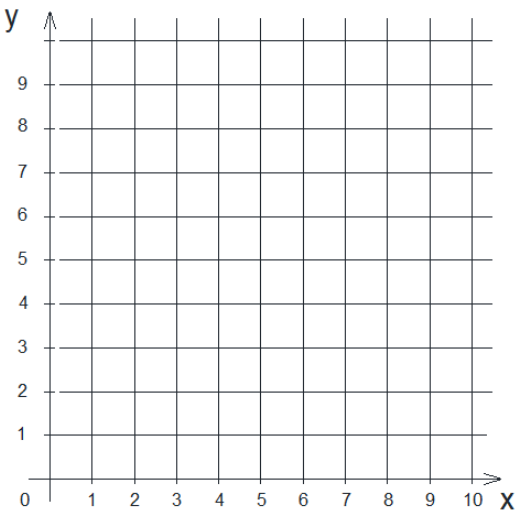

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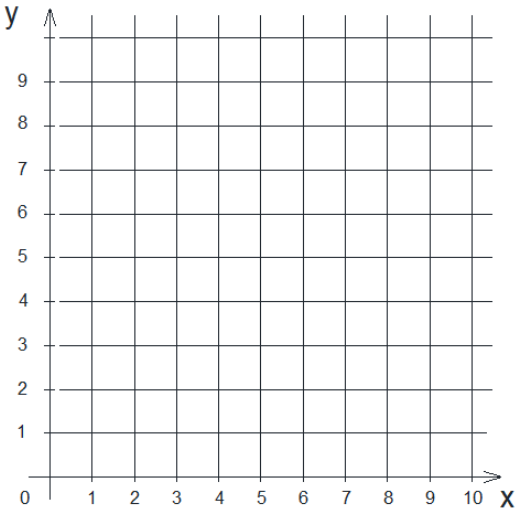

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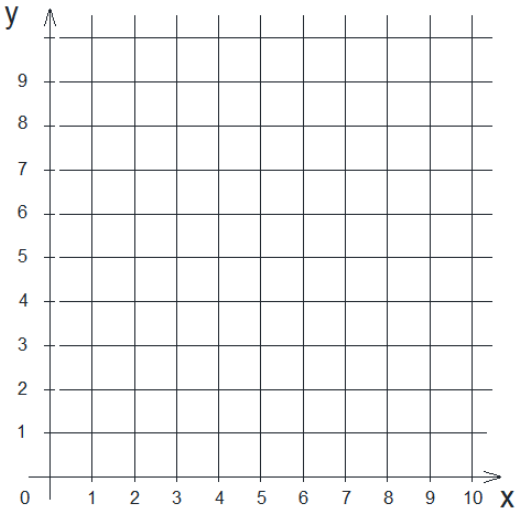

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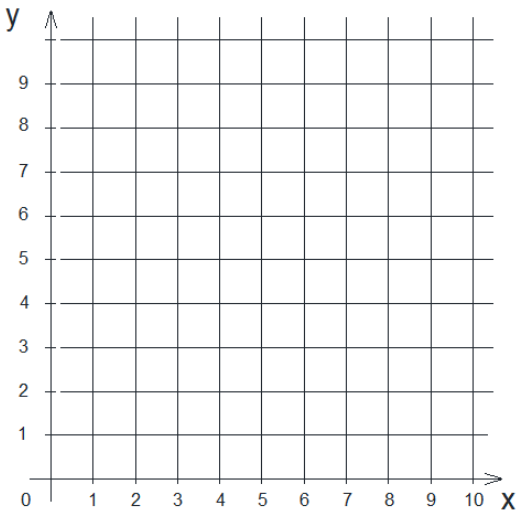

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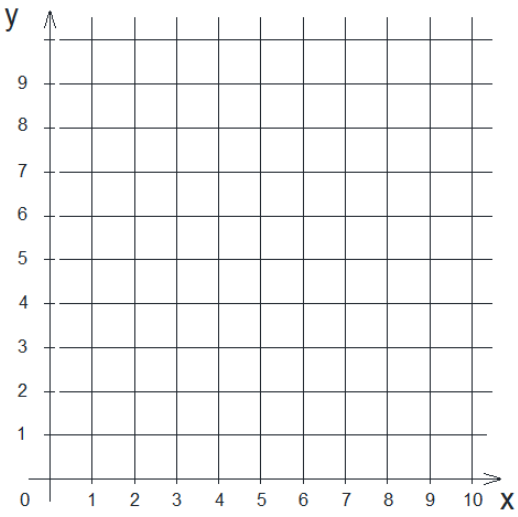

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1 2 3 4 5 6 7 8 9 10
x


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0
1 2 3 4 5 6 7 8 9 10
x





Exercise 6.2

1 Solve the following equations.

a $4x+3=31$ **b** $8x+42=2$

c $6x-1=53$ **d** $7x-4=-66$

e $9y+7=52$ **f** $11n-19=102$

g $12q-7=14$ **h** $206t+3=106$

i $\frac{2x+1}{3}=8$ **j** $\frac{2x}{3}+1=8$

k $\frac{3}{5}x+11=21$ **l** $\frac{x+3}{2}=x$

m $\frac{2x-1}{3}=3x$ **n** $\frac{3x}{2}+5=2x$

2 Solve the following equations.

a $12x+1=7x+11$ **b** $6x+1=7x+11$ **c** $6y+1=3y-8$

d $11x+1=12-4x$ **e** $8-8p=9-9p$ **f** $\frac{1}{2}x-7=\frac{1}{4}x+8$

UNIT FOUR: PROBABILITY AND STADISTICS

EXPLANATION VIDEO:

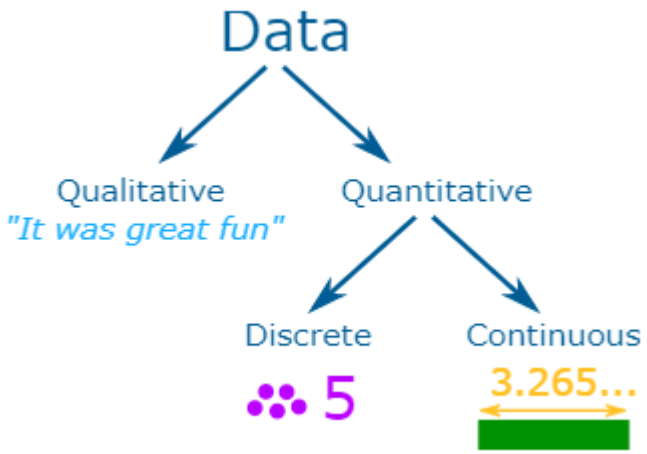
https://www.youtube.com/watch?v=hcgThf5mv38&ab_channel=mathantics

DATA:

Data is a collection of facts, such as numbers, words, measurements, observations or just descriptions of things.

Data can be qualitative or quantitative.

- Qualitative data is descriptive information (it describes something)
- Quantitative data is numerical information (numbers)



Quantitative data can be Discrete or Continuous:

- Discrete data can only take certain values (like whole numbers)
- Continuous data can take any value (within a range)

Put simply: Discrete data is counted; Continuous data is measured.

Frequency Distribution

Frequency

Frequency is how often something occurs.

Example: Sam played football on:



- Saturday Morning,
- Saturday Afternoon
- Thursday Afternoon

The frequency was 2 on Saturday, 1 on Thursday and 3 for the whole week.

Types of data and methods of collecting data

- Primary data – collected by the person doing the investigation.
- Secondary data – collected and stored by someone else (and accessed for an investigation).
- Data can be collected by experiment, measurement, observation or carrying out a survey.

Ways of organising and displaying data

Score	1	2	3	4	5	6
Frequency	3	4	3	5	2	3

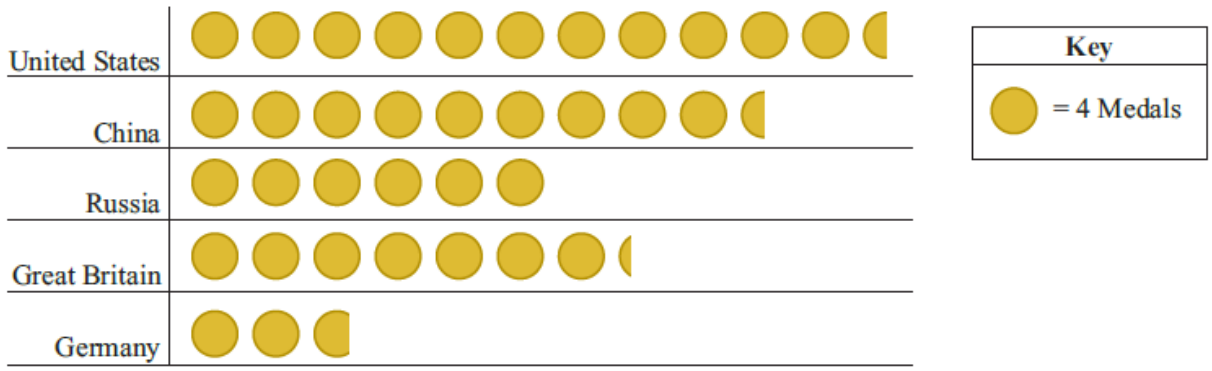
Frequency table (ungrouped data)

Amount spent ($)	Frequency
0 – 9.99	34
10 – 10.99	12
20 – 19.99	16
30 – 29.99	9

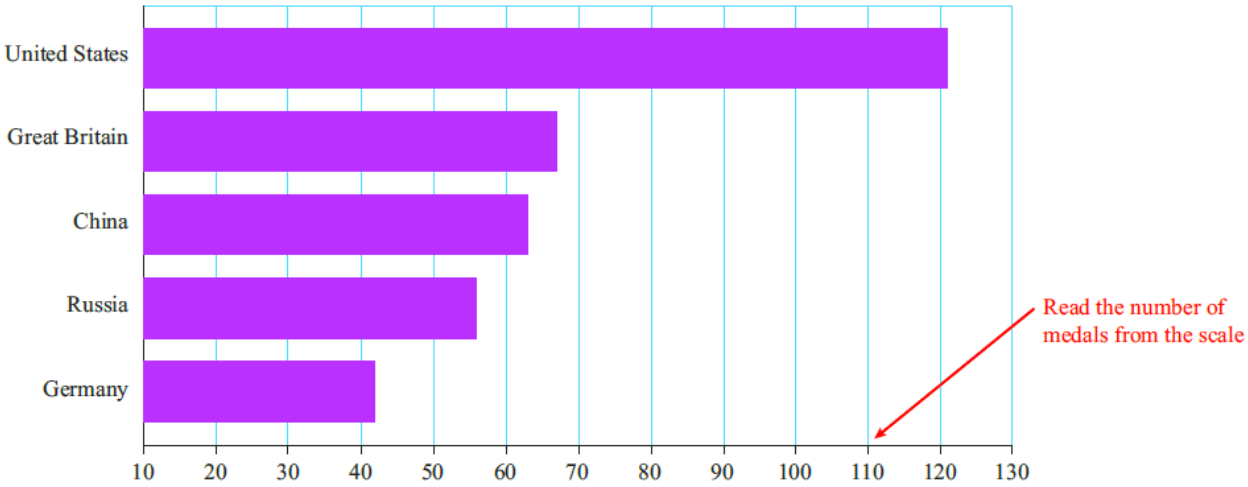
Number of data in that group, not individual values.

Class intervals are equal and should not overlap.

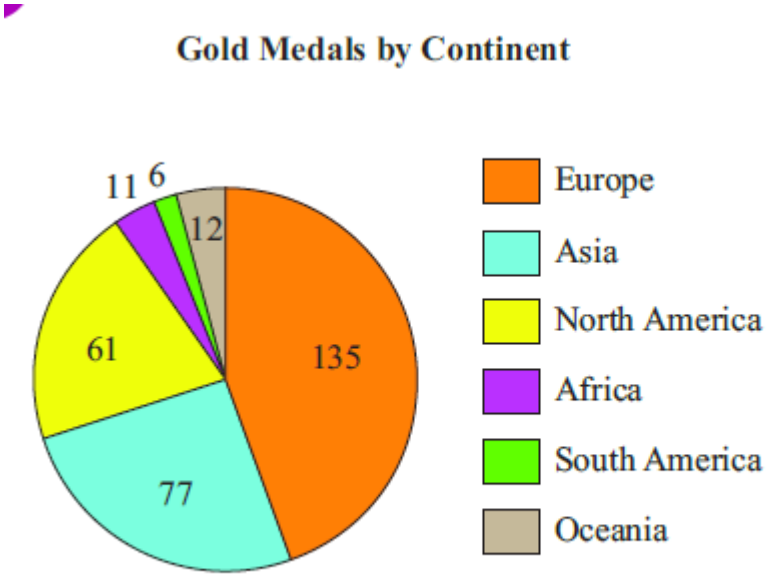
Frequency table (grouped data)



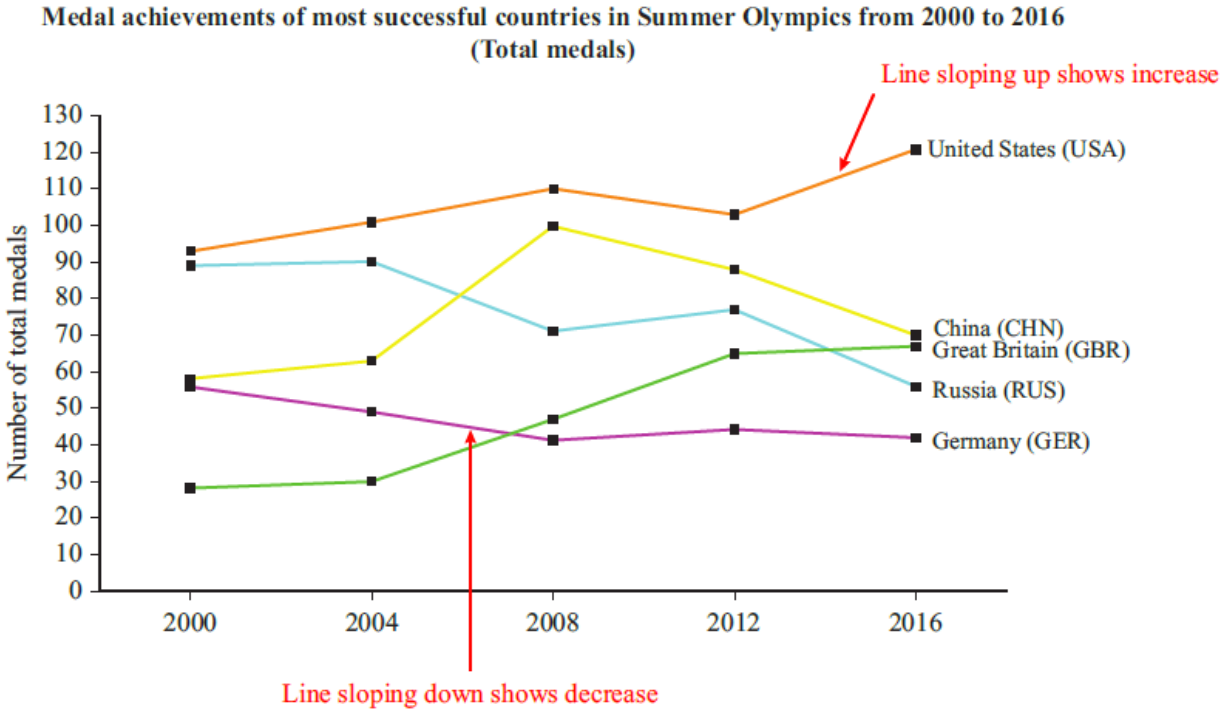
Pictogram – used mostly for visual appeal and effect



Bar charts – useful for discrete data in categories



Pie charts – useful for comparing categories in the data set



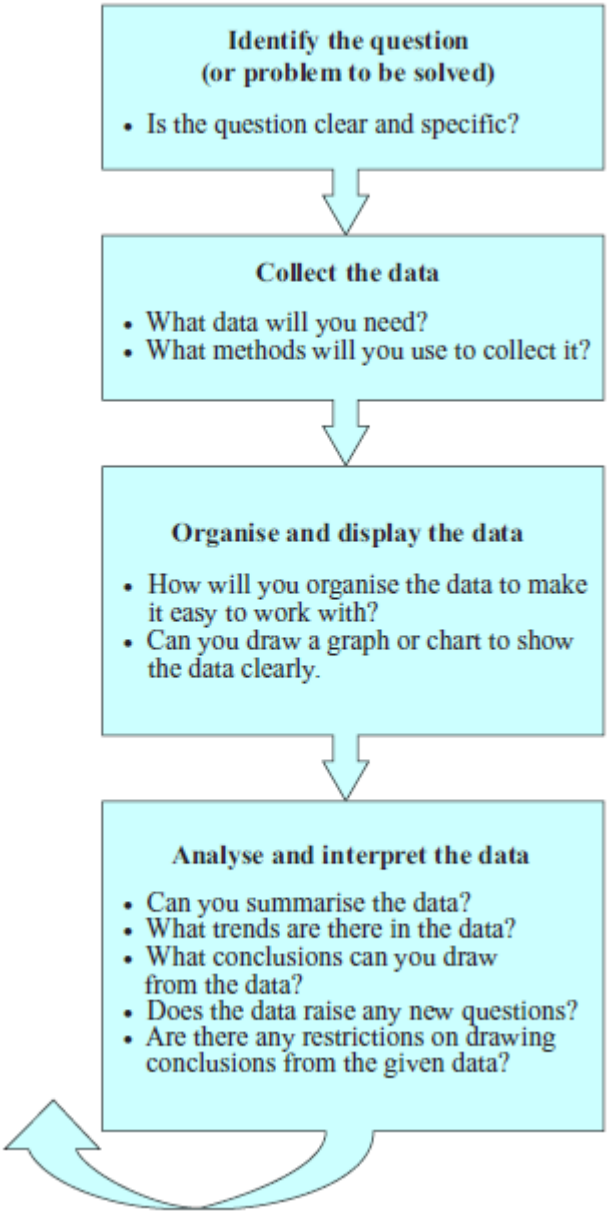
Line graphs – useful for numerical data that shows changes over time

Graphs can be misleading. When you look at a graph think about:

- The scale. The frequency axis should start at 0, it should not be exaggerated and it should be clearly labelled. Intervals between numbers should be the same.
- How it is drawn. Bars or sections of a pie chart that are 3-dimensional can make some parts look bigger than others and give the wrong impression of the data.

Collecting and classifying data

Data is a set of facts, numbers or other information. Statistics involves a process of collecting data and using it to try and answer a question. The flow diagram shows the four main steps involved in this process of statistical investigation:



Different types of data

Answer these two questions:

- Who is your favourite singer?
- How many brothers and sisters do you have?

Your answer to the first question will be the name of a person. Your answer to the second question will be a number. Both the name and the number are types of data.

Categorical data is non-numerical data. It names or describes something without reference to number or size. Colours, names of people and places, yes and no answers, opinions and choices are all categorical. Categorical data is also called **qualitative** data.

Numerical data is data in number form. It can be an amount, a measurement, a time or a score. Numerical data is also called **quantitative** data (from the word quantity).

Numerical data can be further divided into two groups:

- **discrete data** – this is data that can only take certain values, for example, the number of children in a class, goals scored in a match or red cars passing a point. When you count things, you are collecting discrete data.
- **continuous data** – this is data that could take any value between two given values, for example, the height of a person who is between 1.5 m and 1.6 m tall could be 1.5 m, 1.57 m, 1.5793 m, 1.5793421 m or any other value between 1.5 m and 1.6 m depending on the degree of accuracy used. Heights, masses, distances and temperatures are all examples of continuous data. Continuous data is normally collected by measuring.

Methods of collecting data

Data can be collected from primary sources by doing surveys or interviews, by asking people to complete questionnaires, by doing experiments or by counting and measuring. Data from primary sources is known as **primary data**.

Data can also be collected from secondary sources. This involves using existing data to find the information you need. For example, if you use data from an internet site or even from these pages to help answer a question, to you this is a secondary source. Data from secondary sources is known as **secondary data**.

1 Copy this table into your book.

Categorical data	Numerical data
Hair colour	*Number of brothers and sisters*

a Add five examples of categorical data and five examples of numerical data that could be collected about each student in your class.
b Look at the numerical examples in your table. Circle the ones that will give discrete data.

2 State whether the following data would be discrete or continuous.

a Mass of each animal in a herd.
b Number of animals per household.
c Time taken to travel to school.
d Volume of water evaporating from a dam.
e Number of correct answers in a spelling test.
f Distance people travel to work.
g Foot length of each student in a class.
h Shoe size of each student in a class.
i Head circumference of newborn babies.
j Number of children per family.
k Number of TV programmes watched in the last month.
l Number of cars passing a zebra crossing per hour.

3 For each of the following questions state:

i one method you could use to collect the data
ii whether the source of the data is primary or secondary
iii whether the data is categorical or numerical
iv If the data is numerical, state whether it is discrete or continuous.

a How many times will you get a six if you throw a dice 100 times?
b Which is the most popular TV programme among your classmates?
c What are the lengths of the ten longest rivers in the world?

d What is the favourite sport of students in your school?
e How many books are taken out per week from the local library?
f Is it more expensive to drive to work than to use public transport?
g Is there a connection between shoe size and height?
h What is the most popular colour of car?
i What is the batting average of the national cricket team this season?
j How many pieces of fruit do you eat in a week?

4324
343
128
144
4
729
9
512
169
196
216
125
162
1
36
81
64
4
3
2
1
A
B
C
D